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From gameplay to comic book.

A generative artificial intelligence retelling tool
that converts voice-recorded TTRPG sessions into
a multimodal retelling format.

MASTER THESIS

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Klagenfurt, September 22, 2025

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Abstract

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On the usage of Artificial intelligence

In this day and age where Artificial Intelligence is taking the world by storm and many people are using it in their daily lives, and also to write academic works it is important to ask ourselves if and how we should make use of models like ChatGPT, Claude, GitHub co-pilot and other models. Personally I think that the usage of AI within academia should be part of everyone's workflow as sciences should be about discovering new concept and new information. Thus information which a Large language Model can not have, as it does not exist yet. It can also increase the speed of researching allowing for more science to be done. In this thesis ChatGPT-4o and ChatGPT-5 has been used to assist with writing, coding, research, brainstorming and Latex formatting.

For writing it has always been given a personally written piece of text and the model was then asked to give feedback on the work. This created a nice back and forth where the chatGPT acted as a professor who could instantly give feedback.

For coding purposes the model was used after creating the minimal viable product. I first wanted to create my own fully functioning pipeline and only then did I want to create more complex structure within the code. It also helped a lot with creating many different concept prototypes which could be tested quickly, aiding the iterative design process. In the end most code is a combination of ChatGPT-4o and myself, but with the foundations created by myself.

There are a couple ways one could research with AI. You could let the AI do all the work for you, but I think this is a very unethical and useless way of using AI. What do you learn from that? Instead I used AI as an advanced search engine which I can ask to give to find papers for me which I can then read and use. It is also a great tool to reverse search as I learned a lot over the years, but do not necessarily remember all the names of the papers or authors. Asking ChatGPT to find papers based on certain concept and descriptions is a great and fast way to continue your work.

Whenever I was brainstorming I would pitch my ideas to chatGPT to see what it thought about it and to gather feedback on the ideas. ChatGPT was used as something of a peer when no peers where available to talk with.

Lastly, too save a lot of time I often used chatGPT to create latex tables and other formatting needs like lists and greater structures which I can then use to write inside of. For example: I would first write a chapter structure down in a plain text doc and then ask ChatGPT to convert in into a proper latex formatted document.

I think it is important that people do not lose track of how they use AI. It is an important tool, but it should not take over the creating and thought process. Humans should think for themselves and discover novel concept on their own. AI is a great tool but it can quickly fill in a lot of the blanks, resulting in a lack of creative thought. I think this can reduce the capabilities of humans and thus AI needs to be used meaningfully.

List of Acronyms

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Chapter 1

Introduction

1.1 Motivation and Objectives

Storytelling has a rich history of development starting back as far as humanity can go back. From Neandertals 40.000 years ago making cave drawings in an attempt to share experience and knowledge and 5000 years ago the ancient Egyptians made Hieroglyphics to tell stories and record history. To now YouTubers sharing their experience and people on TikTok are recording and sharing everything they can think of. The medium of storytelling has been innovated time and time again, cave drawings, Hieroglyphics, oral, written texts, printing press, photo, film, and everything now in the digital age. But where are we going next? In a world that is ever-changing, super time hungry and where people are looking for more and more ways to consume and share stories. What is the next form of storytelling? What are new forms of storytelling we can use with new technology? As we are at the advent of a great generative artificial intelligence revolution we should look into the possibilities of AI. How can we use generative AI to create the next form of storytelling?

This thesis aims to look into the possible applicability of generative artificial intelligence and locally inferred large language models in combination with game retelling. It will do this by audio recording Table Top Roleplaying Game (TTRPG) play sessions and using generative artificial intelligence to generate a comic book based on the audio recording of the players. Giving the players a new way of sharing their experience, a new way of retelling.

Retelling games in combination with artificial intelligence is a new field of research with a small research demand [23]. However with all the newest, easy to work with and well performing opensource generative models like Stable Diffusion [44] and Llama [50] there is a lot of potential for artificial intelligence as a reteller of games. As we can now generate images and text based on prompts and other input (like a voice recording) we can start to use artificial intelligence as a new form of storytelling. The printing press of the AI era.

This thesis started because of a small research demand for this topic in the academic world [23] and a direct request from my supervisor. For my motivation, I see a lot of potential in the usage of AI as a reteller of games. Primarily for video games, because you can track everything the player does and use that for generating. Besides video games, storytelling games like Dungeons and Dragons [51] and other TTRPG systems can also make great use of such a system. By recording a session and converting it with AI there is potential. TTRPGs are already a big part of the retelling on the internet through digital media like Twitch, YouTube and every podcasting platform out there. One of my favourite Dungeons and Dragons podcasts, The Adventure Zone, have converted their adventure into a comic book [37] and Critical Role even converted their story into a TV show [43]. Clearly there is a want for retelling TTRPG adventures. It is also a medium that leverages the usage of AI properly as there are infinite possibilities while playing TTRPG games and every group makes their own story. Thus the requirement for generative artificial intelligence goes up as there is no possible way of making a system that can incorporate everything people can do in a TTRPG game without generative AI.

The central research question that guides this thesis is:

What are the key design and implementation challenges of a multi-modal generative AI tool for converting play sessions into multimodal retellings?

This question will be explored by developing a prototype system, evaluating its performance, and reflecting on the design process and user experience. Ultimately, the goal is to contribute to the emerging intersection of generative AI, storytelling, and games, while exploring how artificial intelligence might help reimagine one of our oldest cultural practices.

1.2 Structure

The basic outline of the research approach of this master thesis is to create an artefact which converts audio-recorded TTRPG play sessions into a comic book. The methodology that will be used is the Design Science Research methodology [14]. This methodology fits nicely with the research goal as it focuses on creating solutions through iterative development with a focus on testing the artefact. It also gives clear steps for both the design process and for a structured literature review which have been used as the skeleton of the thesis. The structure of the thesis is as follows:

- Chapter 2 discusses the theoretical background and related work, including literature on game retelling, AI-based storytelling, and comic generation.
- Chapter 3 outlines the research methodology and design science framework used to develop and evaluate the artefact.

- Chapter 4 describes the development of the artefact: a pipeline that converts recorded TTRPG sessions into comic books using Whisper, LLMs, and image generation models.
- Chapter 5 presents the evaluation of the system, based on qualitative and/or quantitative testing.
- Chapter 6 offers conclusions, discusses the broader implications of the project, and suggests directions for future research.

Together, these chapters aim to provide a comprehensive exploration of the potential for generative AI to support and expand the ways we retell game experiences.

Chapter 2

Theoretical Research and Related Work

This chapter creates a theoretical foundation to be used throughout the entire thesis. It starts with the concept of retelling games, what it is, why it is important and how it is done. It then discusses comic books as a medium and certain aspects of comic books that are important to understand when creating a comic book. It then looks at different AI models which are required for the artefact that was created for this thesis, with some help of AI. It is followed by ethical consideration of using AI and aspects to consider when working with AI. Followed by best practices for prompt engineering. Lastly it looks at the three primary forms of player interaction in TTRPGs as those are an important foundational aspect for the data gathering of this thesis.

2.1 Retelling games

This section explores the theoretical background of retelling games. It tries to define what retelling is, how narrative, media studies, and game studies can help us understand it, and why retellings are meaningful in both creative and analytical contexts.

Retelling a definition

Game retelling is the act of reconstructing a past play experience into a new form, whether through storytelling, video, illustration, or another medium. It is a way of making sense of what happened during play, translating fleeting moments into shareable narratives. In analogue games like TTRPGs, where much of the story unfolds through improvisation and player interaction, retelling can serve many purposes. It preserves memories, communicates the experience to others, and allows for reflection on the themes, decisions, and dynamics that emerged. Whether casually recounted around a table, written into detailed session reports, or adapted

into comics and podcasts, game retellings are always interpretive. They emphasize certain moments over others, shift the tone or point of view, and reshape nonlinear, messy play into something structured and communicable.

Retelling is not unique to games. It is a deeply human practice, as old as humanity itself. Across history, people have retold experiences and stories to remember, explain, warn, reinterpret, or entertain. Whether through oral traditions, books, historical records, films, podcasts, or personal anecdotes, the act of retelling transforms raw experience into structured narrative [48]. Game retelling continues this tradition: it is a way of making meaning out of play, translating fleeting moments into lasting stories that can be shared, reshaped, and remembered.

Retelling a game is not simply an act of preservation; it is an act of authorship. The moment one begins to retell, decisions must be made: what events matter, which moments carry meaning, and how they should be presented. These choices reshape the original experience, transforming play into narrative. In this sense, retelling becomes a creative process of selection, remixing, and reinterpretation. It is a form of participatory media production that aligns with what Jenkins calls convergence culture: a context in which stories circulate across platforms and audiences actively contribute to their expansion and transformation [28].

Entertaining a retelling of a game is different than retelling a game. The one entertaining it is a passive observer, a voyeur of the story, while the reteller is still an active participant, picking one branch out of the branching narrative to retell in whatever way they see fit. The reteller keeps a level of agency which they can use to see the result of their decisions and choices.

Theoretical Foundation

The retelling of a game is a beautiful example of the difference between cybertexts and traditional literature [1]. In cybertexts, the reader is not just a reader, but a player. They are not just consuming a linear narrative, but they are shaping and discovering an interactive narrative through their effort and interaction with the text. While a retelling of a game is an experience where you as the reader are a voyeur, a passive observer of a story once created by the players. A retelling is not a pure reconstruction of the original play experience, but rather a paratext in the form of a new interpretation of it [24]. The reteller selects, emphasizes, and sometimes even alters events to create a coherent narrative that captures the essence of the original play while making it accessible to an audience that was not present during the game.

In a retelling of a game, much of the original experience is inevitably lost. The original play experience is not just a story, but a complex interplay of rules, player agency, open-endedness, speculation, creativity, and player interactions. We are going from an interactive experience to an immersive one [46], sacrificing agency in favour of structured narrative delivery. The act of retelling removes most of the branching narrative which is essential to gameplay and converts it into a linear or

semi-linear retelling. We lose a level of narrative simulation which is entrusted to us while using more ergodic texts [1].

A retelling of games is often a half-real experience that leans toward fictional narrative [31]. The reteller aims to create a coherent story that can be understood as a standalone artifact, without requiring the audience to grasp the game’s underlying rules. Yet those rules are precisely what shaped the original story. The fact that a player failed disastrously due to a bad dice roll, or triumphed in a hopeless situation through sheer luck, is not just incidental, it is central to the experience. Weaving rule-based elements into the fictional retelling is essential for capturing the full texture of the original play experience.

Recapping is also a form of retelling. Many ongoing TV shows, series, anime, comic books, etc. tend to start each episode with a retelling of the previous episodes, focusing only on the important aspects and leaving out a lot of the scenes. This is done in an attempt to reengage the watcher/reader and to remind them of the flow of the story so that they understand the story better. The same counts for TTRPG groups who are part of a larger story that lasts numerous sessions separated by time. This is done in an attempt to restructure the story and make it clear to all the players to understand where they are in the story. Every time a story is retold it can ascribe new symbolic value through the way it is retold, as the paratext is variable between retellings [24].

Eladhari introduces the concept of game retelling as the fourth layer of narrative [18]. According to Eladhari, this layer exists beyond the embedded, emergent, and interpreted narratives that typically characterize game storytelling. The fourth layer is concerned with post-play reflection and reinterpretation. It allows players, designers, or critics to re-engage with a game experience after the fact, often through a different medium [18]. In this view, retelling is not just about conveying what happened in the game, it is also about creating space for analysis, critique, or emotional processing.

Sych expands on this concept by arguing that critical game retellings can serve as a bridge between personal experience and broader thematic exploration. By combining Eladhari’s fourth layer with the idea of “breaking the fourth wall” Sych shows how retellings can expose the mechanics, social dynamics, and ethical tensions of gameplay itself [49]. This is particularly relevant in TTRPGs, where storytelling and rules are deeply intertwined, and where players often make decisions that reflect not just in-game motivations, but also real-world values and group norms.

Moreover, game retelling in this form invites consideration of what is lost and what is gained in translation. Elements like humor, tension, or spontaneity may be flattened or reshaped, while the visual format offers new tools, framing, symbolism, and visual rhythm to convey meaning. As Eladhari notes, retelling is not just representational; it is generative. It produces new readings, new interpretations, and sometimes even new stories.

Finally, comic-based retelling also offers a lens into how different types of player interaction, such as roleplay, exposition, and rule engagement, manifest in a visual

medium. The project thus becomes a way to explore not only how games are retold, but also how different aspects of a game are retold by artificial intelligence.

2.1.1 Examples of TTRPG retellings

Many people post their TTRPG sessions on the internet as podcasts and/or as video content. These could be seen as retellings as they are most of the time modified versions of the original, however there are some of these shows that go above and beyond and convert their original recording into a true retelling. In this section we look at some of these examples to ground the concept of retelling TTRPGs as something that people want, but also to learn from the retellings and as a source of inspiration.

Critical Role, one of the largest TTRPG podcast shows out there [8], converted their adventures into *The Legends of Vox Machina*[10], a polished animated retelling of their podcast episodes. It is heavily scripted as it truly turns a chaotic TTRPG session into a smooth TV experience. They also created two comic books from their work: *Critical Role: Vox Machina Origins*[12] and *Critical Role: The Mighty Nein Origins*[9]. These are, however, primarily focused on new information and function as a prequel.

The Adventure Zone is a TTRPG podcast [36] that retold their original work as a comic book series [37]. The comics are a beautiful combination of a standalone series, but keep many meta elements of *Dungeons and Dragons* inside of it. The Dungeon Master sometimes makes a cameo in a separate panel where they interact with the world or comment on the world, giving the person that would not have a prominent and obvious role a place to shine. Through this it also stays close to the source material and does not try to change the original story too much. The comic book series is a retelling of the podcast, but it is also a retelling of the game that was played, having both narrative and game mechanics retold.

2.2 Comics as a Medium for Retelling

Creating comic books is an art form on its own, and to understand how to generate a comic book with artificial intelligence, we need to understand how comic books function as an artifact: how they are made, how they are structured, how they are read, and what comic books are. This section will look at comic book studies to create a better understanding of comic books as an artifact.

Defining Comics

Just like how Scott McCloud starts his book *Understanding Comics* we should also ask ourselves: What even is a comic book? Luckily we can just take the definition that McCloud gives us: "A comic book is a juxtaposed pictorial and other images in

deliberate sequence, intended to convey information and/or to produce an aesthetic response in the viewer." [35]. This definition is very complex and for the purpose of this thesis we can simplify it to sequential art [16]. When you take a comic book panel and look at it outside of the context of other panels it is just an image. But when you place them side by side in a sequential order the pictures become a comic book 2.1. Looking at the image we can see that an image of a man holding its hat is just a man holding a hat. But when you place two images of a man holding his hat side by side we can see that it creates a motion of tipping his hat. The panels create an effect of change, an effect of time passing. The space between the panels, knows as the "gutter", create a concept of time in stationary images. Placing images in a sequential manner is the essence of comic books.

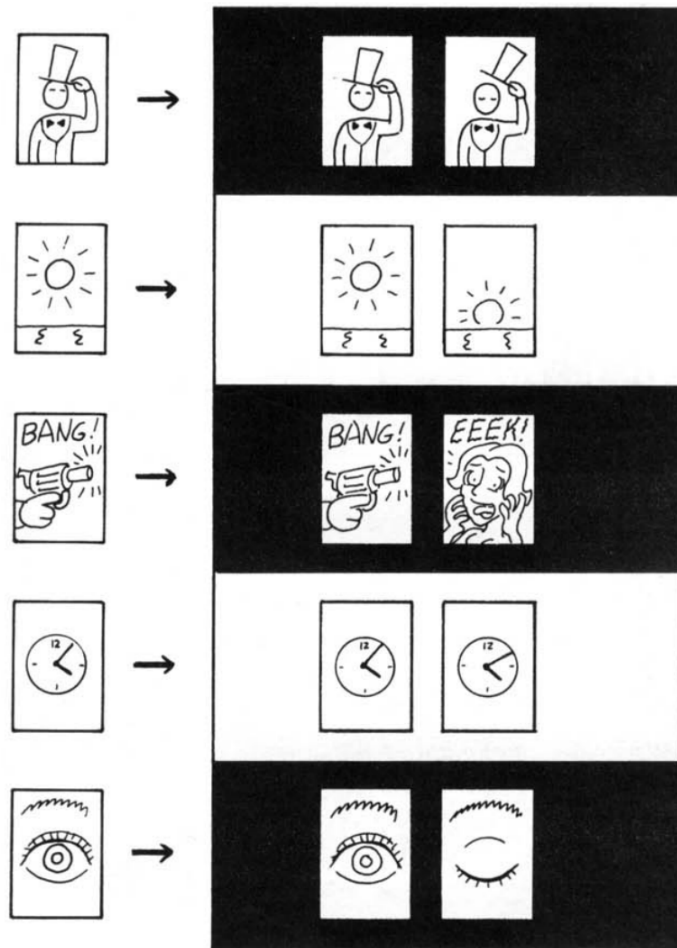


Figure 2.1: Sequential Art page 5 [35]

Comic Book Formats and Reading Conventions

Comic books exist in a wide variety of formats and traditions, each with its own structural and cultural conventions. When people think of comic books, they often picture brightly coloured superhero stories or comedic adventures they read as children. While these are certainly part of the medium’s history, they represent only one branch of a much broader spectrum of comic book formats.

One of the most widely known formats is Japanese *manga*. Manga is typically published in black and white, uses a right-to-left reading order, and often follows a clear panel structure with high variation in panel size and pacing. Manga tends to place a strong emphasis on emotion, movement, and stylized facial expressions, and often incorporates visual manpu symbols to represent feelings or actions.

American comics, on the other hand, are usually published in full color and follow a left-to-right reading order. They tend to use grid-based layouts, with clearly delineated panels and speech bubbles. While the most globally recognized genres in American comics are superhero stories, the format is used for everything from noir and horror to memoir and historical fiction.

Korean *manhwa*, and especially digital webtoons, represent another major tradition. These are primarily designed for vertical scrolling on digital devices, often in full color. Unlike page-based comics, webtoons use elongated vertical panels and spacing to control pacing and emotional beats. This format has become increasingly popular globally, thanks to the easy accessibility of webcomics on the internet.

Beyond these formats, there are countless experimental and hybrid approaches. European *bande dessinée*, Chinese *lianhuanhua*, and graphic novels each bring their own conventions. Some comics, such as Frank Miller’s Sin City, use high-contrast black-and-white art to create a noir aesthetic [39]. Others, like those illustrated by David Finch [20], are known for hyper-detailed linework and realism. Comics are an art form and thus infinite ways of representing them exist.

These different comic book formats are not just stylistic choices, they influence how stories are told, how information is revealed, and how readers interpret visual cues. For any system that seeks to generate comics, understanding these formats is crucial. The chosen structure will affect panel composition, pacing, and even the kind of narrative that can be effectively conveyed.

Comic Book Panels

One of the core aspects of comics is the panel structure chosen by the artist. Some comics even break conventional panel structure and use whole pages as panels, place panels in panels, have panels that are not rectangular, but irregular shapes such as curved, circular, or angled layouts. Some do not even use panels at all but make use of emptiness or use the same character on a long panel with multiple instances of the same character creating a concept of time as you read left to right. Comic

storytelling embraces a wide variety of panelling styles and they shape the comic book.

In between each panel there is a space of nothingness known as the gutter. This is the place where two panels transition from two separate images to one idea [35]. Thought just an empty white space on a page we as humans fill in this empty space with what would/could happen to go from the one to the next panel. The reader becomes an active participant in the story they fill in the gaps. This location of transition has been classified by McCloud into 6 different forms of transition 2.2. This however is not a hard science, but it gives us a better understanding of comics and it can be useful when analysing the narrative structure of the generated comic.

A Moment-to-Moment transition is a small change. On the first panel the person has their eyes open and on the next their eyes are closed. On the first panel a hand is on a table and on the next panel the hand is 2 inches to the left on the table.

An Action-to-Action transition is a more impactful and instant change. In the first panel a person stands ready to hit a baseball with his bat and in the second panel they hit the ball.

A Subject-To-Subject transition is a change of subject while staying within the scene. Switching a panel from one person to another person who are talking with each other. Switching between someone trying to defuse a bomb and the timer on the bomb. These type of panels require the reader to exert more effort to understand what is happening.

A Scene-to-Scene transition is a change of scene which transitions across significant time and space. The first panel is a kid running out of the front door of his house and the next panel being the kid entering the school front door. Two panels showing two people thinking about each other while both sitting on their own bed in their own house. These type of panels require even more effort from the reader.

An Aspect-to-Aspect transition is different as it bypasses time and focuses on different aspects of a place, a mood or an idea. Showing a number of panels which together create the scene. Create understanding of someone enjoying their holiday by showing them sunbath, followed by a panel of the ocean followed by a panel of the sky with the sun. These type of transitions anchor the reader into the world.

Lastly Non-Sequitur transitions are a transition that offers no logical relationship at all. Switching from a boy making his math exam to a painting of a chicken to the sun to a white wall. There is no underlying connection between these panels.

Besides the transitions of panels there is also an importance on how many transitions and panel one uses to convey an idea. If a comic uses more panels to convey an idea the scene will most likely become more vivid for the reader as there is a lot more context. If there are too little panels there might even be too little context for the reader to understand what is happening. More panels could thus be valuable if you try to visualize a very intense and important moment.

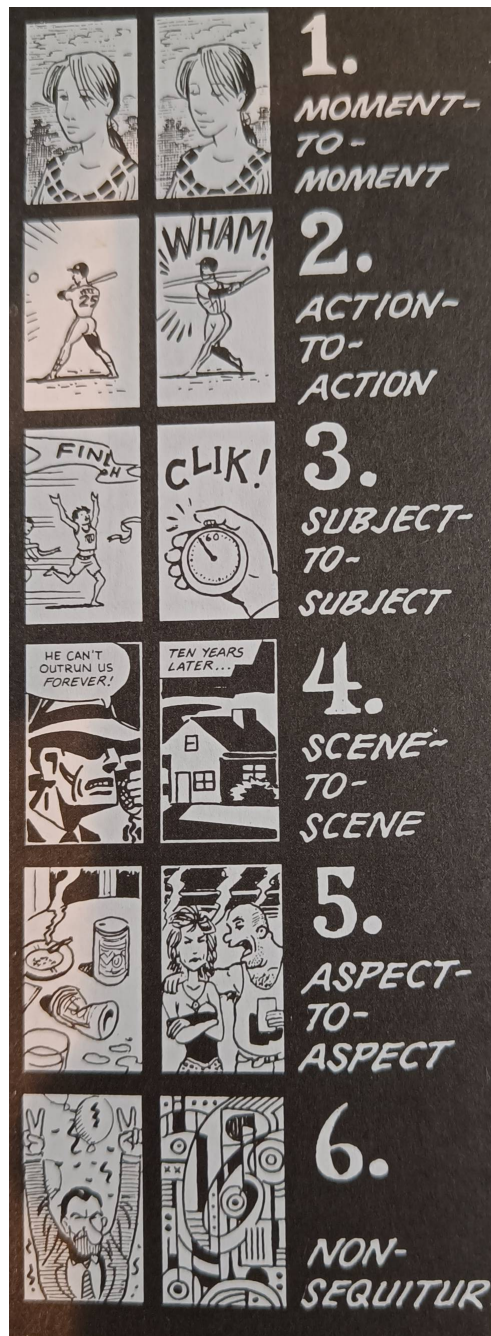


Figure 2.2: 6 forms of panel transitions [35] page 74

Text, iconography and speech bubbles

Text in comic books is one of the primary ways to convey information to the reader. It is used to convey dialogue, thoughts, sound effects and narration. The way text is presented can greatly affect the reading experience. Just like while writing this thesis I have to use *italics* to emphasize a word, use **bold** almost exclusively for headings, and I could use underlines but that is no where to be found in this text. Comics also make use of these different text styles to convey different meanings. Use bold to put emphasize on something, use italics to indicate something is fast or to indicate a thought, and use underlines to indicate something is important and there are way more forms of typographic emphasis than these three and also way more ways they are being used.

Iconography is essential to many comic books to communicate those things that can not be easily spoken or written down. In the world of manga we call this unspoken communication *Manpu* 2.3. To quote Matt Alt *Manpu "speak" what can't be easily spoken (or written) in words* [32]. "ZZZ" to indicate someone is snoring or sleeping, a circle with stars and birds above a characters head indicate a state of unconsciousness by an impact of sorts, pages being torn of a calendar indicate time passing and one of my favourite making an L with your thumb and index finger and placing it under you chin to indicate deep thought (or the one I do not like where it represent people acting cool).

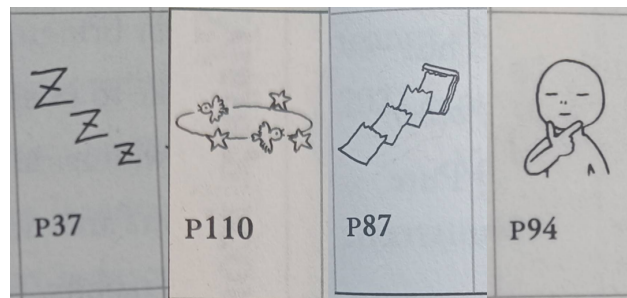


Figure 2.3: Iconography [32]

Text is also used to indicate sound that is part of a panel. The sound of a knife chopping vegetables is often represented by the text "chop chop chop". The sound of a gunshot is often represented by the text "bang" or "pow". The frequency, amount, size, typographic emphasis all have different effect on this use case of words creating different meanings with different or the same words. These sounds are not just there to indicate what is happening, but also to create an atmosphere and to make the reader feel like they are part of the story.

2.3 Generative AI Models for Multimodal Output

There are many different AI models floating around the internet that could be used as part of the artefact that was made for this thesis. To choice which model to use this chapter talks about different models and why they were chosen for this thesis. In the end Llama 3.1-8B has been chosen as the large language model, Stable Diffusion XL has been chosen for the image generation and Whisper has been chosen for the audio transcription model. The primary reason for these choices is that they can be run locally, have good enough performance and have decent online documentation.

2.3.1 Large Language Models

Picking a suitable Large Language Model for the project is a quite the impossible task as models keep changing rapidly and new models are released every day. There are many academic papers out there discussing models and their performance, usage and more [2, 11, 26, 52]. However all of these are already more than a year old and as models are changing so fast most of those papers are already outdated, however still very useful. As this paper is not an analysis of AI models but a practical thesis I have asked ChatGPT-4-turbo to give me a list of models with their pros and cons and asked it to make me a table for Latex with the information. The prompts that I used: *I would like to have a table with a comparison of the top 10 large language models. Show the pro's and cons of the models. Add if they are open source and locally runnable.* Followed by: *Can you convert that table to Latex code?.* The results can be seen in table 2.1, the latex code had to be modified to fit the page.

Even here we can see that the models are out of date. Gemini 1.5 Pro is not the latest model as Gemini 2.5 is already released, and LLaMA 3.1 is not the latest model as LLaMA 4 is already released. However, the table does give a good overview of the models that are available and their pros and cons. Primarily the question if they are open source and can be run locally. I used this method not because I see it as a academic valuable method but because I want to use technology and see how useful it could be for academia. Looking at this small experiment we can see that it could be useful, however the data is very outdated and not very useful. It does point us in the correct direction and gives us a good overview of the models that are available.

Looking at the options provided we can quickly cut out most of the models as they are not open source and there is no possibility of running them locally. The models that are left are LLaMA, Mixtral and Phi. Out of these LLama is the best option for this artifact. It provides a state of the art model with regular updates and a large community behind it. Furthermore I have worked with Llama in the past and I know how to work with it.

Llama has many different models to pick from, ranging from 7B to 70B parameters. The larger the model the better the performance, however the more compute it requires. As this project is done on a consumer GPU (NVIDIA RTX 3080 Ti) we

Model Name	Developer	Pros	Cons	Open Source	Locally Runnable
GPT-4o	OpenAI	Multimodal (text, image, audio); fast and versatile	Occasional hallucinations; high compute needed	No	No
Claude 3.5 Sonnet	Anthropic	Safety-focused; handles long contexts; natural responses	Slower; closed API only	No	No
Gemini 1.5 Pro	Google DeepMind	Strong multimodal support; integrates Google ecosystem	Resource-heavy; limited open variants	No	No
LLaMA 3.1	Meta AI	Open-source; good at translation and code	Requires technical expertise; energy-intensive	Yes	Yes
Mistral Large 2	Mistral AI	Strong reasoning and code generation; long contexts	Commercial license; proprietary	No	Limited (license)
DeepSeek-R1	DeepSeek	Multilingual support; efficient	Limited adoption; less documented	No	No
Mixtral	Mistral AI	Efficient Mixture of Experts; open-source	Shorter context; niche usage	Yes	Yes
Grok 3	xAI	Good social media integration; open-source focus	Requires subscription; limited distribution	Yes	Possibly (limited)
Command R+	Cohere	Up-to-date info retrieval; good conversational skills	Limited multimodal support	No	No
Phi-2	Microsoft	Open-source; efficient size/performance	Limited multimodal support; some bias concerns	Yes	Yes

Table 2.1: Comparison table of 10 LLM’s by ChatGPT-4-turbo

are limited to the smaller models. So the only option is the Llama 3.1-8B model. This model provides good enough performance for our use case and can be run on a consumer GPU. All the other models are either too small for the functionality that is required or too large for the available hardware.

2.3.2 Image generation models

To pick a suitable image generation model I have used the same method as with the large language models as those have the same issue of changing very fast [5]. I asked ChatGPT-4-turbo to give me a list of the top 10 image generation models with their pros and cons and if they are open source and locally runnable. The results can be seen in table, the latex code had to be modified to fit the page. The prompts that I used: *I would like to have a table with a comparison of the top 10 image generation models. Show the pro’s and cons of the models. Add if they are open source and locally runnable.* Followed by: *Can you convert that table to Latex code?*

In this table there are 5 models that could be used, however 3 of them have the issue of having a small community or little documentation. The models that are left are Stable Diffusion XL and DeepFloyd IF. Stable Diffusion XL is the most popular model and has a large community behind it, making it easy to find documentation and support. DeepFloyd IF is a model primarily used for photorealistic images and has a strong prompt fidelity, however it is more GPU intensive and slower than Stable Diffusion XL. As we have to generate a comic book with multiple images, Stable Diffusion XL is the best choice for this project as it is the most flexible and has the best community support. Also as we are generating many images we need a model that can run on a consumer GPU at a decent speed, which Stable Diffusion XL can do. Lastly DeepFloyd IF focuses on photorealistic images, which is not what

we are looking for in this project as we want to create a comic book with a more stylized look.

Model Name	Developer	Pros	Cons	Open Source	Locally Runnable
DALL-E 3	OpenAI	Strong text understanding; detailed coherent results; good at stylized scenes	Closed-source; watermarking; only accessible via API	No	No
Stable Diffusion XL (SDXL)	Stability AI	High-quality output; flexible prompting; large community support; LoRA compatible	Slower on consumer GPUs; inconsistent hands/faces; no native inpainting	Yes	Yes
Midjourney v6	Midjourney Inc.	Excellent aesthetics; strong stylization and detail	Only available via Discord; lacks fine control; closed-source	No	No
Kandinsky 3.0	Sberbank	Multilingual; hybrid CLIP/VAE approach; decent composition	Smaller community; less documentation and support	Yes	Yes
DeepFloyd IF	Stability AI	Photorealistic images; strong prompt fidelity	Multi-stage pipeline; GPU intensive; slow inference	Yes	Yes
Playground v2.5	Playground AI	Good people rendering; open weights; simple setup	Limited documentation; fewer tools than SDXL	Yes	Yes
Muse	Google	Token-based fast decoding; high prompt accuracy	Not released to public; re-search only	No	No
Runway Gen-2	Runway ML	Supports both image and video generation; intuitive UI	Closed-source; limited free use; lacks local control	No	No
Imagen	Google	Extremely high fidelity; excellent text-image alignment	Never released; purely re-search/internal	No	No
PixArt-a	Alibaba DAMO	Fast generation; supports style text control; efficient	Limited English documentation; smaller ecosystem	Yes	Yes

Table 2.2: Comparison table of 10 image generation models by ChatGPT-4-turbo

2.4 AI Ethics in Creative Applications

This section discusses the ethical consideration when creating an AI system based on the ethical guidelines for trustworthy AI given by the European union [19]. This document consist of 4 Ethical principles which should be followed and 7 key requirements which any AI system should follow. The goal of this section is to understand what aspects to keep in mind during development, but also to have a ruler to measure the final artifact against to see if the resulting artifact is in line with European guidelines and if not why that is the case.

2.4.1 4 Ethical Principles - Introduction

The ethics guidelines for trustworthy AI gives us 4 ethical principles which should be followed to create an ethical and just AI system. These AI ethics are grounded in the fundamental rights within the EU treaties, the EU Charter and the international human rights law. Furthermore these guidelines are overarching guidelines based partly on a more difference set of principles proposed by the European Group on Ethics in Science and New Technologies[22]. These guidelines are however not without their issues and might have tension between them where the designer will have to make an acceptable ethical trade-off for their system, if possible. Some of

the aspects of the ethical principles have been left out as those are way out of scope for this project and academic research in general.

The principle of respect for human autonomy

This principle discusses the idea that human beings need to have freedom and autonomy and that an AI system should not interfere with these rights. The goal of AI systems should be to augment, complement and empower human beings in aspects of life. And AI systems should be there to support humans to create meaningful work which can contribute to the world. Because of this the design of AI systems should be done from human-centric design principles, resulting in humans to still take the meaningful and potential impactful opportunities up to humans. Humans should always have oversight over AI work processes. Lastly any human being using an AI system should have full and effective self-determination over themselves.

The principle of prevention of harm

This principle is in short the first law of Robotics from Asimov: A robot may not injure a human being or allow a human to come to harm through inaction. In further detail the principle says that an AI system should not harm or otherwise adversely affect human beings. This is not only about physical safety, but also about human dignity and mental health. To ensure this a system should be technically robust and not open to malicious use. The extent of prevention of harm does not stop with humans, as it should consider the natural environment and all living beings. Lastly vulnerable persons and asymmetries of power or information should receive particular attention when designing AI systems.

The principle of fairness

This principle discusses the idea that development, deployment and use of AI systems should be fair and if possible should create more fairness within the world. The prevention of bias, discrimination and stigmatization should always be a core part of any AI system. Fairness also extends to equal opportunity in terms of access to AI technology, educations, goods and services and people working in the sphere should foster this. Most importantly is that AI systems should never deceive or unjustifiably impair people of their freedom of choice. Because of this the decision-making processes should be identifiable and explainable by humans.

The principle of explicability

This principle discusses the explicability of AI systems and their components as this is crucial for building and maintaining trust in AI systems. In essence this means that a system should be transparent, openly communicated about its capabilities and purpose, the output of the system should be explainable. Sometimes there are

AI systems which can not be explained and those require special attention. In these cases special attention should be paid to explicability and other ways of explaining the system should be considered, such as traceability, auditability and transparent communication on system capabilities. Explicability is however in a weird position as it is not very concrete and can differ widely in how much is required depending on the AI system and its context.

2.4.2 Realizing Trustworthy AI 7 requirements

This section discusses the 7 key requirements which AI systems to follow to be become trustworthy AI systems. These requirements are build upon the 4 ethical guidelines. Furthermore they are accompanied by technical and non-technical methods to achieve this goal.

Human agency and oversight

In essence all AI systems should support human autonomy and decision-making for this the first step is that the system protects the fundamental rights of people. In cases where this is a risk, a fundamental rights impact assessment should be done. Furthermore users of the AI system should be able to supply feedback in case the fundament rights might be infringed. The second step is to give users the knowledge and tools to understand and interact with the AI system in such a way that they can assess the system and challenge the system. Ai systems should further the capability of humans and support them achieve the goals they want to achieve. The focus should be on improving human autonomy and thus it should avoid (incidental) manipulation, deception, herding and conditioning. Lastly to achieve human autonomy, the system should have enough oversight capabilities so that humans can override any decisions made by the system.

Technical robustness and safety

Trustworthy AI systems should have a certain degree of technical robustness. This requires that during the development of an AI system to extra care is taken around risk mitigation and prevention to such an extent that the system reliably behave in the way it is designed to behave. It should also have a robust security system in place to protect itself and its data against outside threats. AI systems should have fall back options so that if the system fails to function there are other systems or humans that can be used to take its role. The system should also have a high level of accuracy in terms of what it outputs and in cases where occasional inaccurate predictions can not be prevented it should be made clear that this is the case and how likely this is to happen. Lastly AI systems should have a high level of reproducibility as well as reliability.

Privacy and data governance

As with all IT systems that use data it is of vital importance that privacy of data is guaranteed and in accordance with the GDPR. Thus data needs to be handled with care and not used in any unlawful manner or used in an unfairly discriminating manner. Data that is used to train AI should also be addressed before it is used as the integrity of the data needs to be ensured.

Transparency

Within the domain of transparency there are 3 topics, traceability, explainability and communication. Traceability means that the data that is used to train models should be documented to the best possible standards and the way the AI system makes decisions is documented. This all to increase transparency, which allows developers to see why a system could create wrong outputs. Explainability entails to being able to explain why an AI system functions in the way it does and why it is used in certain contexts. Lastly communication entails that the user of an AI system is aware that it is using an AI system and not for example talking with an AI chatbot instead of a human customer support agent. Furthermore the end user should understand what the capabilities and limitations are of a AI system.

Diversity, non-discrimination and fairness

This requirement focuses on the inclusion and diversity throughout AI system's their life cycles, from design to development to operation. The diversification of AI systems should be supported by the unbiased datasets and diverse development teams with different backgrounds, cultures and disciplines. Identifiable bias in datasets should be removed during the data collection and a diverse development team should be garnered during the hiring process. Furthermore AI systems should be accessible to all people and should not behold a one-size-fits-all approach, but instead understand that there are different people with different needs and means. Lastly it is important to discuss the AI system with stakeholders to gather feedback on the system so that it can become a more fair system.

Environmental and societal well-being

AI systems should think about the impact they can have on the broader society, other sentient beings and the environment. The entire life cycle of an AI system should be considered in term of energy consumption and resource usage with the goal to have a less bad effect on the environment. The social impact as a result of certain AI systems should also be considered as certain systems can have impact on how people behave and think. Lastly AI systems should not threat upon the democratic process of society or other societal perspectives.

Accountability

The accountability requirement ties all the requirements together by placing accountability on the creators of AI systems and their outcomes. To comply with this an AI system, its data and the design process should have a high enough level of auditability so that the trustworthiness of the system could be tested. In case the trustworthiness is in question there should be mechanisms that allow people to report on these issue in a safe manner with the overarching goal to minimize negative impact from AI systems.

When working with the requirements stated above tensions between them is bound to appear and in such a situation trade-offs need to be made. These trade-offs should be considered carefully in a rational and methodological manner. This way if any problems arise it can be documenter ad used to better the system in the future. This way an AI system can garner most trust from its users and become a better system in the end.

2.5 Best practices for prompt engineering

Prompting any langue model is a kin to instructing a human, however the machine will take everything you say without context and thus prompting is a skill on its own. We can it prompt engineering for a reason after all. This section goes over important aspects that you should keep in mind while prompting based on articles of creators of different AI models. Specifically meta their guide to Llama [38] and OpenAI's guide to their models [41]. Furthermore some guide that are floating around the internet for Stable Diffusion are discussed [3, 4]. Lastly a one article is used to string everything together [17].

2.5.1 What is prompt engineering?

Prompt engineering is the act of writing, refining, designing, and optimizing an instruction (the prompt) for a large language model. Resulting in a effective way of interacting with the model. The goal is to communicate the user's intent in such a way that both the machine and user have the same idea as to what needs to happen. This process is an essential step when working with LLM's if you want to obtain accurate, relevant and coherent responses. As LLM's are becoming better and more prevalent in our daily life understanding how to do prompt engineering will become a critical skill for anyone wanting to make full use of LLM's.

2.5.2 Large language model prompting

The prompting of large language models has become a whole discipline in the world and I assume it will become part of many people their daily life and in academia it will become part of courses and research. This section however will go into what the

best practices are when instructing LLM's based on the instructions of the creators and articles on the web.

Be clear and concise

A prompt should be straight to the point and easy to understand so that it can generate the output that you are looking for. It should provide just information for the model to understand what to do and it shouldn't consist of too much information as the model might get confused from a larger context. Lastly it is important to avoid jargon and technical terms as the model might confuse those words with the wrong meaning.

Use specific examples

When you are looking for specific output from the model it is important to provide an example of what the output should look like. Thus if you want a specific JSON format output you should provide that specific output once so that the model can work with the example and follow your instruction better.

Test and refine

Just like engineering and writing, it is all about iteration. Testing the prompt you write and refining it over and over again will result in a better result in the end. This is also the best way to learn how to work with prompts and will make you a better prompt engineer in the long run. You should start with a zero-shot prompting approach, this is where you give the model the bare minimum of instructions and see what the results are. If this is not enough you can move to few-shot prompting which is where you provide a couple of examples of what you want. And if that is still not enough you can move to fine-tuning the model.

Role based prompts

If you are looking for the model to respond in a certain way, you can instruct the model to embody a specific role. The role can be anything from a university professor or a student to a tour guide or a tech support. This way the model understands the context it is operating from better and it understands the perspective of the person interacting with the model better allowing for a more tailored response.

Chain of Thought technique

The chain of thought technique is a way to describe to the language model how it should think about and respond to the question it is being asked in a specific order. First do X, then do Y, then do Z, etc. This way the LLM knows exactly what to do and it can easily follow your instructions.

Reduce fluff and imprecise descriptions

Do not leave your instruction ambiguous. If you tell the model to give a few sentences, it will not be very consistent and it might not be in line with what you thought. Say 3 to 5 sentences instead. Avoiding fluff also makes it so that your instructions are clearer to both yourself and the language model. Resulting in a better results.

Say what to do, not what not to do

A large language model cuts your instructions up into pieces and then converts it to number making a guess for what to do respond with. Thus if you say something like: Do not ask for someone's age. It might lose the context of the not part and ask the age of someone. Because of this is it better to be precise about what the model should do and to avoid things that it shouldn't do.

Tone and style

The tone and style you write your prompts in dictates the tone and style the model will respond in. If you ask very academic question in a very academic way it will respond in an academic manner. If you ask questions in a more free form and/or childish form it will respond in kind.

Leading words

If you are using the model for very specific tasks such as code or prompt generation it is important to words that are linked to that. So if you want code for a specific project, mention the coding language you use and the way you want to run it. For example if making a web application with python, you should mention that you are using Python and which libraries you want to use such.

Prompt order

The order you write your prompt in an important way to make sure that the model understand what you are asking of it. A prompt can have many parts such as: Task, example, role, goal, etc. Starting with the role, followed by the task, then the goal and then the example is better then doing it in reverse. What the exact best way of ordering a prompt is has not been found, however it is a important consideration to play with when prompt engineering.

Using explicit constraint

Using explicit constraints in a prompt results in the LLM having to do less guess work and thus can generate better results that are in line with what you want. Some

examples of things you can constrain are: which data to use, format of the output, scope, and length of the output.

Balancing user intent and model creativity

Instructing everything to the last letter what the LLM has to do will most likely result in failure of the prompt. LLM's need a bit of space to what they are good at and to showcase their creative strength. Thus balancing the user intent and the model creativity is an important aspect of prompt engineering and one which should be kept in mind when trying to generate very specific results.

2.5.3 Image generation model prompting

The creating of prompts for image generative models is to and extent very similar to prompt engineering any LLM as they are both text based instruction models. Most of the aspects of making a good LLM prompt are transferable to image generation prompts, however there are some aspects unique to image generation which are discussed in this section.

Keywords

One of the important aspects of a good image generative prompt is the usage of keywords. These are specific words which you do not place in the model, but are categories which the model uses in the background. Examples are subject, medium, style, resolution, color, and lightning. These are important categories for you to describe so that the model understands what you are looking to generate.

Negative prompt

Image generative models have a system to allow the user to describe that which they do not want to see in the image. This is a great way to steer the image being generated as you can get more specific on the image you want to create from two different angles. Writing a negative prompt can be done in the same way one writes the primary prompt, however most of the time a negative prompt consists of styles and attributes such as disfigured deformed, ugly, and low resolution. But you can also do more specific things like saying magic if you are generating a wizard.

Association effect

Models are build to link certain words with other words and thus specific words are associated to other words. This can have a large impact on the image you are generating as for example specificizing that you want someone with blue eyes will most likely result in the model generating a Europeans, Caucasian with blue eyes. While if instruct the model for brown eyes you will get a more diverse sample of

people. This also happens with country names, celebrities, artists, and way more. This is an inherent bias that is inside the data used to train models and thus is part of the output as well. Being aware that this association effect exists and thus could impact the result of your generation is vital to generating images.

Custom models

Most image models are general models which generate anything the user asks of them, however if the user wants to generate a specific art style, pose, person, etc. without having to describe that they can use a fine-tuned models or fine-tune their own. Fine tuning allows a user to feed images that are similar to the result they are looking for into a model to train the model in what they should generate. For example a user could insert a certain comic book into a model to teach it to only create images that are similar to that art style. This is very beneficial if one wants to make many images with similar features or is looking to leverage models in more consistent ways. Examples of custom model types are checkpoint models, textual inversion, LoRA, LyCORIS and hypernetwork.

Using visual UI's

Almost all image generative models can be run from code using either an API call or running it locally on you computer, however there are also some visual user interfaces which can be used to generate images and these have many benefits. AUTOMATIC1111 is one of the best and most used image generation UI which is very powerful. It removes many of the issues models have and gives many handholds for the user to generate custom images. Loading in custom models, overcoming token limits, and more are all build in the system. However it can not be used to automate image generation (Outside of scripting) as that requires the usage of code.

Image refiner

Some model also have an extra model that can be used after generating the image. A so called refiner model which will clean up the generated image [?]. Make it sharper, more vibrant, up-scale the resolution, denoise the image, etc. This is an image to image model which can be used as an extra step, but is not required. It does take a lot longer than the generating of the image as it requires more function evaluations.

2.6 Three primary forms of player interaction

This section discusses the idea that TTRPGs have three primary forms of player interaction: **roleplay, exposition, and rule interaction**. The purpose of intro-

ducing this concept here is to establish a framework that can later be used to gather data in Chapter 6.

Roleplay is where the players, including the Game Master, speak to each other while acting as their game characters, or in the case of the Game Master, as one or more NPCs. This kind of in-game interaction aligns closely with the narrativist aspect of GNS theory, where the goal is to create a story together [15]. It is also referred to as *in-character talk* and is one of the core communication styles during play [6, 21].

Exposition is when the Game Master narrates a moment in time, a location, or something else the players are experiencing in the game. The goal is to create a vivid image that helps the players immerse themselves in the world. Exposition sits between the narrativist and simulationist aspects of GNS theory and is generally classified as *out-of-character speech*, since it describes situations rather than presenting dialogue between characters [6]. Its function is to construct a virtual reality in which the players can act, making exposition the key mechanism for creating these worlds [45].

Rule interaction is when the mechanical aspects of the game come into focus. Rolling dice, drawing cards, making checks, and using abilities are central to this form of interaction. By nature, it leans toward the gamist side of GNS theory, though depending on the system it can also overlap with narrativist or simulationist play. At its core, rule interaction is about using the system’s mechanics to overcome challenges. From a communication perspective, this fits into *meta-talk*, though the boundary is often blurry [6]. Discussing rules, probabilities, or possible outcomes clearly falls under meta-talk, yet players may also minimize this by embedding mechanics into roleplay or exposition. This illustrates what Juul calls the *half-real* nature of games: the interplay between rules and fiction, where meta-talk brings the game’s virtual reality to life [30].

Chapter 3

Research Methodology

In this chapter, the research methodology used in this thesis is outlined¹. The chapter begins by introducing the overarching framework of *Computing as a Discipline* and how it informs the design and evaluation of the artifact. It then discusses the Design Science Research methodology, which provides a structured approach to creating and evaluating the artifact. The chapter concludes with a detailed description of the data collection and analysis methods used to evaluate the system, including case studies, self-observation, and expert reviews.

3.1 Computing as a Discipline

This thesis adopts the framework of *Computing as a Discipline* [13] as its primary research approach. This framework enables a systematic and scientific investigation centered on the development of digital artifacts. Within this framework, the design paradigm is used, which consists of four iterative steps:

1. **State requirements** - Identify the goals, constraints, and desired outcomes of the system.
2. **State specifications** - Translate the requirements into measurable and testable criteria.
3. **Design and implement the system** - Develop the architecture and build the working prototype or tool.
4. **Test the system** - Evaluate the system against the specifications through testing or case studies.

In Step 1, *State requirements*, this thesis addresses the need to define design requirements for a system that uses artificial intelligence to retell gaming sessions.

¹As per advice of my supervisor, this chapter has been heavily based on the works of Mia Mohač [40]

TTRPGs are chosen as the basis due to their rich narrative structure and the absence of a complex digital system. This choice reduces the system’s scope and complexity, focusing on story-based retellings instead of interactive gameplay or video capture.

In Step 2, *State specifications*, the system’s requirements are translated into a set of functional and non-functional specifications that guide the design and implementation.

The functional specifications are as follows:

- The system must transcribe audio from a single-microphone recording of a TTRPG session.
- It must generate a narrative script based on the transcribed text.
- It must generate visual comic panels that align with the script’s events and dialogue.
- It must compile the script and images into a comic book format suitable for reading.

The non-functional specifications include:

- The output must be understandable without prior knowledge of the session.
- The system must run fully automatically, with no manual artistic or editorial intervention.
- The comic must reflect key narrative moments, character presence, and setting changes.

The system is scoped to retell sessions from tabletop role-playing games and does not attempt to represent mechanics, dice rolls, or real-time interactions. Its focus lies in retelling emergent stories visually, using artificial intelligence as a generative tool.

In Step 3, *Design and implement the system*, the artifact is developed using an iterative design process aligned with the Design Science Research (DSR) methodology. Before conducting any evaluation, the full AI-driven pipeline must be constructed. This pipeline includes modules for transcription, script generation, and image generation. The specifics of this development process are detailed in the next section.

In Step 4, *Test the system*, the complete pipeline is evaluated in a case study by converting multiple TTRPG session recordings into comic books. No manual corrections or tweaks are made to the output, as the goal is not to fine-tune a specific model but to assess the pipeline holistically. The aim is to uncover design requirements that inform future development of AI-based game retelling tools.

To ensure consistency during testing, a single Game Master runs the same TTRPG session multiple times with different groups of players. This creates a controlled environment while still allowing for natural variation. The results are cross-referenced to observe how the system handles minor differences between groups, pacing, and content across recordings.

Participants provide both qualitative and quantitative feedback via semi-structured interviews after reviewing the generated comic. Additional expert feedback is collected from multiple GMs who evaluate the system’s output and usability. This multi-perspective evaluation helps identify strengths, weaknesses, and areas for improvement in the artifact.

3.2 Design Science Research as a Methodology

The Design Science Research (DSR) methodology is a way to create an artifact while adhering to scientific research methodology[14]. It is a systematic approach that emphasizes the creation and evaluation of innovative artifacts to solve real-world problems, primarily used within the engineering disciplines. As this thesis aims to create a novel artifact and as I am deeply familiar with the DSR methodology, it will be the primary methodology used in this thesis. There are different versions/interpretations of DSR, but the one used in this thesis is structured around 7 different guidelines[27] which can be seen in Table 3.1.

Guideline	Description
Guideline 1: Design as an Artifact	Design-science research must produce a viable artifact in the form of a construct, a model, a method, or an instantiation.
Guideline 2: Problem Relevance	The objective of design-science research is to develop technology-based solutions to important and relevant business problems.
Guideline 3: Design Evaluation	The utility, quality, and efficacy of a design artifact must be rigorously demonstrated via well-executed evaluation methods.
Guideline 4: Research Contributions	Effective design-science research must provide clear and verifiable contributions in the areas of the design artifact, design foundations, and/or design methodologies.
Guideline 5: Research Rigor	Design-science research relies upon the application of rigorous methods in both the construction and evaluation of the design artifact.
Guideline 6: Design as a Search Process	The search for an effective artifact requires utilizing available means to reach desired ends while satisfying laws in the problem environment.
Guideline 7: Communication of Research	Design-science research must be presented effectively both to technology-oriented as well as management-oriented audiences.

Table 3.1: Design-Science Research Guidelines (Copied from [27] page 83)

Guideline 1: Design as an Artifact

The central contribution of this thesis is a working artifact: a multimodal AI pipeline that transforms TTRPG session audio recordings into comic books. This includes transcription, transcript-to-script conversion, image generation, and layout formatting components.

Guideline 2: Problem Relevance

The problem addressed is the lack of AI-based tools for automatically retelling gaming experiences, particularly those rooted in narrative-heavy analogue formats like TTRPGs. This project explores how artificial intelligence can support the automated retelling and archiving of such experiences. The resulting insights aim to inform the design of future game retelling systems and contribute to the broader field of AI-assisted narrative generation.

Guideline 3: Design Evaluation

The system is evaluated through case studies in which the same TTRPG session is run multiple times by the same Game Master but played by different groups. This setup ensures narrative consistency, enabling direct cross-referencing of the resulting comics. The outputs are reviewed by session participants and expert Game Masters, combining qualitative user feedback with functional evaluation to assess the system’s effectiveness.

Guideline 4: Research Contributions

This thesis contributes a novel tool for AI-assisted narrative retelling, along with insights into the design requirements for such systems. It offers a practical demonstration of integrating generative models into a creative pipeline and contributes to ongoing discussions about how games can be represented and retold after play.

Guideline 5: Research Rigor

A rigorous approach is applied to both the design and evaluation of the artifact. The development process follows established software engineering practices, including iterative prototyping, testing, and refinement. The selection and use of AI models are grounded in documented performance and methodological consistency.

Guideline 6: Design as a Search Process

The artifact could be constructed in countless ways by varying models, prompts, and parameters. Rather than attempting exhaustive optimization, this thesis seeks to develop a satisfactory and functional solution from which useful design requirements

can be derived. This allows the project to contribute broadly applicable insights without being constrained to a single ideal configuration.

Guideline 7: Communication of Research

This thesis is written to be accessible to both technical and non-technical audiences. It includes detailed descriptions of the design process, system architecture, and evaluation methodology. In addition, visual examples of the generated comics are included to concretely demonstrate the system’s capabilities and outcomes.

By following these 7 guidelines, this thesis aims to produce a meaningful contribution to the field of AI-assisted narrative generation, while also adhering to the principles of design science research.

3.3 Data Collection Methods

This section discusses the methods used to evaluate the usability and effectiveness of the developed system. A combination of qualitative and quantitative techniques will be employed, including a case study, self-observation, and expert review. These methods aim to uncover strengths and limitations in the system, with a focus on identifying concrete design requirements for a narrative retelling tool in the context of TTRPGs.

Case Study

The first step in the evaluation process involves generating a number of comic books using the system. To do this, a case study will be conducted in which a single Game Master runs the same TTRPG session multiple times with different groups of players. The session will be run with 2 or 3 players and will last between 1 and 2 hours. This session length is chosen to avoid the need for breaks during the session, which should result in less requirement for post-processing and a more streamlined recording. The game used will be *Dungeons and Dragons 5th Edition* as this is a system which is well known by players and it is easy to find participants already familiar with its rules. Furthermore, it is important that the players are knowledgeable about the game, as this minimizes rule-related interruptions, simplifying the session recordings.

The session will be recorded in a manner which is true to how players would normally go about recording a session, using a smartphone. The smartphone used is a Samsung Flip 5, as this is the phone owned by the Game Master. Using the same phone creates consistent recordings, which is important for consistent results. The primary reason as to why a smartphone is used is because it would be uncommon for a random TTRPG group to have a high-quality microphone available. The location of the recording will be in the same private home of the Game Master, as this is a natural location for a TTRPG session to take place. The entire setup has been

chosen to keep the ecological validity of TTRPG sessions as high as possible[33]. By staying true to real-world conditions, the evaluation remains grounded and realistic.

After the sessions have been conducted, the recordings will be processed by the system to generate a comic book. The comic book will then be presented to the players of the session. Participants will be asked to complete a questionnaire containing both closed (quantitative) and open (qualitative) questions, allowing for a well-rounded understanding of how the comic book was received.

Self-Observation

The comic books generated by the system will also be evaluated by the researcher. The purpose of this is to cross-reference the comic books to each other to identify major deviations in output across identical input sessions. As we are using artificial intelligence to generate the comic books, a high degree of variation is expected due to the random nature of AI generation. This will also make it possible to see if the system has issues with certain aspects of a session and if there are consistent problems at certain points of the session. Primary aspects looked at will be the overall narrative, text, characters, art style, completeness, and the overall quality of the comic book. The researcher will also look at the comic books from a technical perspective, assessing system performance in transcription, script generation, and image generation. This process helps identify both strengths and areas for improvement within the pipeline.

The key focus of the self-observation is understanding what makes the output feel, or fail to feel, like a comic book. As the purpose of this thesis is to identify design requirements for a retelling tool, it is important to discover these requirements by cross-referencing the comic books to each other, to other comic books, and to the input of the participants of the case study.

Expert Review

The last form of evaluation for this thesis will be an expert review in the form of a semi-structured interview with multiple Game Masters. The purpose of this is to link the narrative structure and components of a TTRPG session to the comic book. This way, we can see if the important aspects of a TTRPG session are represented in the comic book and if the comic book is a good representation of the session. Furthermore, this will also create a greater understanding of what should be part of a retelling of a game session, as the Game Masters will be able to give feedback on what is missing and what is present in the current work, allowing for a more complete understanding of the design requirements of a retelling tool.

In addition, a semi-structured interview will be conducted with the Game Master who ran the recorded sessions. This provides valuable insights into the intended narrative and supports more accurate self-observation. It also reduces the need for the

researcher to manually parse and cross-reference all session details. This interview also provides a strong reference point to support the self-observation process.

3.4 Data Analysis Methods

This section outlines how the collected data will be analyzed to evaluate the system and extract concrete design requirements for a TTRPG retelling tool. Both qualitative and quantitative analysis techniques will be used to interpret the results from the case study, self-observation, and expert review.

Case Study Analysis

The case study will provide both quantitative and qualitative data through questionnaires completed by players after reviewing the generated comic book. The quantitative data will primarily come from closed-ended questions, which include Likert-scale items evaluating aspects such as clarity, enjoyment, narrative coherence, and visual quality. The responses will be summarized using descriptive statistics, including averages, standard deviation, and frequency distribution. This data will help identify the overall player perspective on the artifact and clarify which aspects of the comic book were particularly successful or in need of improvement.

The qualitative data will come from the open-ended questions in the player questionnaires. The answers given by the participants will be analyzed using thematic analysis. It is expected that the identified themes will inform concrete design requirements for a narrative retelling tool. Thematic analysis will involve coding the responses to identify recurring themes, patterns, and insights related to the comic book's design requirements. This qualitative analysis will provide a deeper understanding of player experiences and expectations, complementing the quantitative findings.

Self-Observation Analysis

A significant portion of the analysis will be carried out by the researcher through self-observation of the comic books generated by the system and the transcripts of the play sessions. This will be done in a structured manner, beginning with a general review of the comics and transcripts while making minor notes. Following this, key aspects such as narrative structure, consistency across sessions, visual quality, character representation, and overall completeness will be assessed. These observations will be documented and compared across the different generated comics to identify patterns in system performance and points of failure.

The self-observation will also include a technical analysis of the system, in which the researcher examines the individual outputs of each stage in the pipeline to identify major issues. This helps determine whether any shortcomings in the final comic

book stem from the input recording, the system’s interpretation, or the image generation stage.

Expert Review Analysis

The interviews with the Game Masters will follow a similar approach to the player questionnaires. The interview will be recorded, transcribed, and coded for a thematic analysis by hand. Common themes, patterns, and insights will be identified to understand how well the comic book represents the TTRPG session and what design requirements emerge from expert perspectives. The analysis will focus on finding design requirements for a retelling tool as per the research question of this thesis. These interviews are expected to provide valuable insights, as the researcher can engage in an open-ended conversation with experts on the topic.

Additionally, the transcripts will be analyzed using an AI-based tool as an exploratory step. While not central to the evaluation, this secondary analysis may reinforce the manual thematic findings and offer insights into the potential of AI-assisted qualitative analysis.

3.5 Ethical Considerations

As this thesis involves human participants who are recorded, it is important to consider the ethical implications of the research. Participants will be informed about the purpose of the study, what participation entails, and how their data will be used. They will be made aware that participation is voluntary and that they may withdraw from the study at any point without any consequences.

The anonymity of the participants will be preserved by not using any personal information in the thesis. Recordings, questionnaires, and all other data will be anonymized and processed exclusively using offline tools. No data will be stored on cloud platforms. Participants will also be informed that they may request the deletion of their data at any time. All collected data will be permanently deleted after the thesis is completed and will not be used for any purpose beyond this research.

Given that this project makes extensive use of artificial intelligence, there is a possibility that participants may be misrepresented by the system. Participants will be explicitly informed that the output is fully AI-generated and that the system is imperfect. They will be warned that the comic book adaptation may diverge from their expectations and that there is a potential for unintentional misrepresentation or emotional discomfort.

Every effort will be made to minimize these risks through transparency, clear communication, and participant agency in the process.

3.6 Limitations

All research methods have their limitations, and this thesis is no different. This section outlines the limitations of the chosen methods and how they may affect the results and generalizability of the findings.

Case Study Limitations

The case study approach used in this thesis has several limitations that should be acknowledged. Firstly, the results may not be generalizable to all TTRPG systems or play experiences. The study focuses on a single TTRPG session, using one system and one Game Master. This significantly reduces variation, which limits the applicability of the findings beyond this specific setup. Secondly, the use of a single Game Master may introduce bias due to their individual play style and narrative structure. Finally, since the sessions were limited in duration and designed as short one-shots, they do not capture the complexity of long-form campaigns, which often involve evolving character arcs and emergent storytelling over extended periods.

Self-Observation Limitations

The self-observation method provides valuable insight into the design requirements for a retelling system, but it is inherently subjective. As the researcher is also the system designer, there is a risk of bias in how issues are identified, interpreted, or prioritized. Although structured note-taking and consistent review criteria are used, the lack of a second reviewer or formal coding system may affect reliability. Furthermore, while the researcher is knowledgeable about games, this expertise may be skewed toward specific types of games, leading to potential blind spots. As a result, certain design requirements—particularly those relevant to unfamiliar game types or retelling styles—may be overlooked. This bias is partially mitigated through expert review, which provides an external perspective on the system’s outputs.

Expert Review Limitations

The expert interviews offer deep, experience-driven feedback, but are limited in scale and diversity. Only a small number of Game Masters are consulted, and their perspectives may not fully represent the wider landscape of game retelling practices. Additionally, as the interviews are semi-structured and conversational, variation in questions and depth of discussion may affect the consistency and comparability of the responses.

The Limitations of Tabletop Role-Playing Games

A broader limitation of this thesis is that Tabletop Role-Playing Games have been chosen as the primary focus. While TTRPGs provide a rich testbed for narrative

retelling, they might limit the generalizability of the findings to other types of games, such as video games and board games. TTRPGs have unique characteristics, such as player-driven narratives and improvisational elements, which may not be present in other gaming formats. As a result, the design requirements identified in this thesis may not be directly transferable to other game formats or retelling contexts. Future research would be needed to explore how these findings extend to other domains.

The Limitations of Artificial Intelligence

As this thesis heavily relies on generative artificial intelligence, the limitations of these technologies also affect the results. The system uses models for transcription, script generation, and image generation, all of which are prone to specific types of error.

The output of these AI systems is inherently non-deterministic, meaning that results can vary significantly between runs even when the input remains the same. This reduces reproducibility and makes it difficult to draw firm conclusions about system performance. Additionally, AI models often introduce biases [34] or hallucinations[29], producing content that may not reflect the original input accurately. This is especially problematic in a retelling context, where preserving the intent, tone, and factual details of the session is important.

Image generation models, in particular, may struggle with maintaining visual consistency across characters or scenes, leading to confusing or misleading representations. Furthermore, these models may reproduce stereotypical or culturally insensitive imagery due to biases in their training data, posing ethical risks that require careful handling.

Together, these system-level limitations constrain the reliability and fidelity of the retelling output, and should be considered when interpreting the results of the evaluation.

Chapter Conclusion

This chapter outlined the methodological approach taken to evaluate the TTRPG retelling system. The study employed a combination of case study research, self-observation, and expert interviews, with both qualitative and quantitative data analysis methods. Key ethical considerations were addressed, particularly in relation to the use of AI and the handling of participant data. Finally, the limitations of the methodology were critically examined, acknowledging constraints in generalizability, system performance, and potential bias. Together, these methods provide a well-rounded and reflective foundation for assessing the design requirements of game retelling tools.

Chapter 4

Artefact Development

This chapter discusses the implementation of the artefact developed for this project. Firstly the system requirements are discussed, followed by the implementation of the artefact itself. The chapter concludes with a discussion of the challenges faced during the implementation and how they were overcome.

4.1 System requirements

The basic system requirements of the artefact is that it can convert a single track audio recording of a TTRPG session into a comic book without any additional user input. Furthermore the system needs to be run on a computer without a connection to the internet. Thus no API calls can be made. To achieve this an iterative design process was used to develop the artefact

Development Environment

Creating complex artificial intelligence agents that need to be locally inferred requires many different libraries and other programs. Because of this a development environment was created to allow for easy development and testing of the artefact. To achieve this WSL2 (Windows Subsystem for Linux) was used to create a Linux environment on a Windows machine. This allows for the use of many different tools and libraries that are not available on Windows. The development environment is based on Ubuntu 24.04.2 LTS and uses Python 3.12.3 as the main programming language.

Running many different AI models requires a lot of resources, especially when it comes to memory and processing power. The development environment was run on a machine with the following specifications:

Nvidia RTX 3080 ti GPU with 12GB of VRAM, Intel i7-9700K CPU with 16 cores and 64GB of RAM. This is my personal machine and was just enough to

handle the models used in this project. Running the models with any other programs running would cause the system to run out of memory and and stop working.

MVP pipeline

The MVP pipeline has 4 steps of processing to go from audio recording to comic book. First Whisper Turbo transcribes the audio recording to raw text. Then Llama 3.1 converts the text to a comic book script. The script is then processed by Stable Diffusion XL to generate the comic book panels. Finally, the panels are assembled into a comic book format using a custom python script. The pipeline is shown in figure 4.1. The audio recording is a single track recording of a tabletop role-playing game session. The prompt used for Llama can be seen in listing 4.1. This prompt generates a description of the panel art and a line of dialogue or narration for each panel. The output is a JSON array of objects, where each object contains the panel art description and the text. The panel art description is then given to Stable Diffusion XL to generate the image with no other instructions. The text is used as a caption for the image. The images are then assembled into a comic book format using a custom python script which generates a PDF file with the images and captions.

Listing 4.1: Prompt for Comic Book Script Generation

```
1 You are given a segment of a Dungeons & Dragons session transcript:
2
3 Session Content:
4 {content}
5
6 Your task is to transform this into a comic book script, broken down into
   individual panels.
7
8 Each panel must include:
9 - A detailed visual description suitable for image generation (key: "panel
   art")
10 - A line of dialogue or narration (key: "text")
11
12 Instructions:
13 - Structure the output as a JSON array of objects.
14 - Each object must contain exactly two keys:
15 - "panel art": A vivid, standalone description of the panel art. Include
   setting, characters, action, and mood.
16 - "text": A single line of spoken dialogue or narration. Rephrase as a
   proper sentence if needed, but keep the meaning true to the original
   session.
17 - Process the entire input and split it into as many panels as necessary.
18 - Do not include any additional explanation, commentary, or text
   outside of the JSON array.
```

```

19
20 Output only a valid JSON array. Example format:
21
22 [
23 {
24     "panel art": "A dimly lit tavern filled with laughing patrons. Magnus,
        a red-bearded human, clinks mugs with Taco and Merle at a wooden
        table.",
25     "text": "So our story begins with the three of you sitting at this
        tavern."
26 },
27 {
28     "panel art": "Close-up of Magnus, his face lit by candlelight, a
        sheepish grin on his face.",
29     "text": "Nobody wants chairs anymore. Everyones into wrought iron now."
30 }
31 ]

```

4.1.1 Lessons learned

The MVP was a success in terms of validating the product idea and gathering valuable knowledge about the product requirements and limitations. In this section lessons learned are described in detail and it is discussed how these lesson will be implemented in the next iteration of the product.

Art

The first thing one might see when looking at the MVP output is the art. The art quality is low, it is not very detailed, the characters are not very expressive and the art style is very inconsistent. There are also a lot of generative AI artifacts, like weird faces and inconsistent human anatomy. Furthermore all these panels are situated in the same tavern, but the there is no consistent setting. All these panels also describe 4 different characters and but there is no consistent character design between panels.

Text

The text has one obvious flaw of clipping outside of the image which is something that needs to be fixed. Other then that when reading the text it is noticeable that it is not really character dialogue, but things the players are saying. Some of the text is player to player dialogue and not in character dialogue which is something that needs to be filtered out. The text also does not make any since in term of story structure. These are just lines of text.

Looking the entire transcription of the tabletop role-playing game session it can quickly be seen that there is no structure to the text, making it extremely difficult for an LLM to make anything meaningful out of it. Figure X is a expert of the transcript where you can see that it is just a consistent stream of text without any structure.

Composition

All the panels have a similar camera angle and composition.

4.1.2 Solutions

In this section possible solutions are discussed which will be implemented for the next iteration of the product. The goal is to improve the quality of the art, text and composition of the comic book panels.

Art

Stable Diffusion is a pre-trained model that is made to work for any art style, to create a consistent art style you could write a art style prompt, however this creates issues with the token limit that stable diffusion has. The proper way to do this is by using a LoRA (Low-Rank Adaptation). LoRAs are a way to fine-tune a pre-trained model on a specific task or domain. In this case, a LoRA can be trained on a specific art style to create a consistent art style for the comic book panels. This will also allow for more detailed and expressive characters. The LoRA can be trained on a dataset of comic book panels in the desired art style. The LoRA can then be used in Stable Diffusion to generate the comic book panels with the desired art style. However since training a LoRA is outside of the scope of this project a pre-trained LoRA from civitai.com will be used. This also comes with the benefit that any LoRA can be used to generate any art style wanted by the user.

Character consistency

To create consistent characters between panels there are 5 different approaches that can be taken. LoRA's can be used for each character creating a very consistent character design. This does require each character to have a LoRA trained on it meaning that you need to have art of the character to start with. This would also mean that the images would be generated with multiple AI's generating images and then having to combine them. The second option would be image-to-image reference chaining. With this method you give SDXL a reference image of the character and then generate the panel with the character art as a reference.

The easiest and least consistent manner would be to use consistent prompting. This means that you give the same prompt for each panel with the character description in it. This will however not create a very consistent character as each

iteration of the prompt will interpret the prompt differently. Resulting in many different smaller inconsistencies. This however would be a good step if the other methods would be too time consuming to implement. It would most likely require two iterations of generating the image, one for the character and one for the panel art. As the number of tokens in the prompt would be too high to generate the panel art and character art in one go.

Inpainting is a technique where you alter part of an image so that it fits your desired effect. You select an area of the image and then fill it in with a new image, generated by AI. This gives you a way to control very specific parts of an image and finetune an image. Normally this requires a lot of manual work where you have to cut out the area which has to be painted in. It would however be possible to use computer vision to detect the character in the image and then use that as a mask for the inpainting. This might result in a very good solution, but there are potential issues with multiple characters in the same panel.

Lastly an option could be using ControlNet. This is a technique where you use a pose reference images to control the pose of the character in the image.

In the end it would most likely be best to use most of the above techniques in combination to create the best possible result. Create a LoRA to have consistent character art, use inpainting to place the art on top of the panel art and use ControlNet to control the pose of the character.

Scene consistency

In the current comic page of the MVP each scene is taking place in the same tavern, but the tavern is not consistent between panels. Most solutions given in the previous section on character consistency could be used here as well, however as there are way more variables possible for locations creating a LoRA is impossible and would defeat the purpose of using AI. You could however let AI create images of a tavern and use those images as reference materials for future uses of that tavern. Meaning you need to keep track of scenes and locations so that you can bring back the same panels in later parts of the comic book.

Text

To fix the text that is in the panels of the comic books the first step would be to fix the transcript of the game session. As is always the case with AI: Garbage in is garbage out. Fixing the transcript will also have effect on the other parts of the comic book, as there will be a clearer structure to the text which can be used in a better way to generate images and create structure between panels. To fix the transcription there are a few that need to be taken.

After transcribing the audio recording with whisper there are 3 steps we can take to increase the usability of the text. First we need to use a Punctuation and Capitalization Restoration model to fix the text by adding punctuation and capitalization to the text. This will make the text more readable and easier to

understand. The second step would be to use Speaker Diarization to identify the different speakers in the text. This will benefit with knowing who said what at which time and allows to have accurate character dialogue. Furthermore this makes it so that the Game Master can be identified as someone who plays multiple characters. A tool to do this is Pyannote Audio, an open-source toolkit written in Python for speaker diarization. Lastly combine the cleaned up whisper transcription with the speaker diarization to create a structured text file.

The next step is to then use an LLM to check all the text and identify what is character to character and what is player to player speech. Creating a clear divide between what does and what does not belong in the comic book. For the game master the same thing needs to happen, but then it is required to identify which character they are playing at that moment. Then an LLM would need to convert the text into real sentences which make sense in the context of the scene as something a someone in the English language would say.

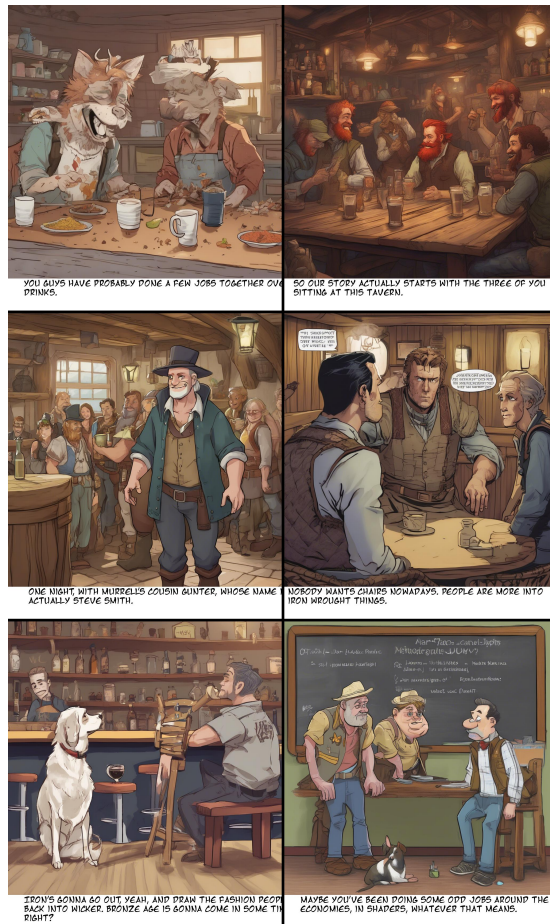


Figure 4.1: The minimal viable product pipeline

4.2 The Artefact

The final artefact that was used to conduct the experiments is a Python program that can be run from the command line. The program takes a single audio file as input and produces a comic book as output. The comic book is generated by using several different AI models that are run locally on the machine. The artefact is designed to be run without any user input, meaning that it can be used to convert a TTRPG session recording into a comic book without any additional work from the user.

To create the artefact for this thesis an iterative design process was followed which was accompanied by a of agile development strategy. To increase chance of success and to heighten usability of the system each part of the system was developed and tested separately. This allowed for many quick changes to be made to the system and to alter the system where needed after testing.

Development started based on a barebones pipeline where there were 3 steps of modification of the input audio. First the recorded audio would be converted to text with an speech to text model. Then the text would be converted to a comic book script with a large language model. And lastly this script would then be turned into a comic book with an image generation model. Creating this system resulted in the minimal viable product which laid the foundation for the final artefact. This first iteration did everything in a one shot approach where all the system did was pas the result of each model to the next. There was no cleaning of data or considerations about the structure of data in-between AI models.

Testing the MVP resulted in many different insights which have been used to create the final pipeline model. The final pipeline takes the recorded audio and converts it into text, then cleans up the text through punctuation and diarization. Resulting in a system that can reconstruct which player said what at what time. The output of this was then handed to a LLM to generate the comic book script in the form of JSON output. Each panel has its own output and the system uses a adaptive sliding context system to allow the system to use as much of the original work as it needs. Afterwards this JSON output was given to an LLM so that it could convert the JSON into a prompt which an image generative model could use to generate images. Then the prompt where given to an image generator using a LORA and lastly the images where placed in a comic book structure.

4.2.1 The final pipeline

In this section the final pipeline is discussed. The pipeline consists of 5 steps which are all discussed in the following subsections. An overview of the entire pipeline can be seen in figure X.

Transcription

For clean transcription of a single audio track that has multiple people talking and sometimes through each other, a number of steps had to be taken to turn the audio into usable data. First off the entire audio recording was transcribed using Whisper. This results in a enormous blob of text with no punctuation and no understandable order of words. Sentences are mixed up, words are missing letters, and way more other issues. To clean up this data two steps had to be taken. First of the audio had to be split up into the different participants. To do this the diarization toolkit pyannote.audio has been used. This AI model looks at the audio and creates a file with time stamps and speaker. Resulting in a file with timestamps that shows which speaker speaks at what time. Combining these timestamps with the whisper output which also has timestamps results in data which clearly shows who said what at what time. Lastly punctuation was restored using the fullstop-punctuation-multilang-large model. This was done so that in the next step it is clear for the LLM what sentences are. This combination of 3 different AI models resulted in clean data which can be used in the next step of the pipeline. The output of this step is a structured JSON file with 4 Key-value pairs which can be seen in listing 4.2.

Listing 4.2: Prompt for Comic Book Script Generation

```
1 {  
2 "start": 0.0,  
3 "end": 2.96,  
4 "speaker": "SPEAKER_02",  
5 "text": "Our story actually starts with the three of you..."  
6 }
```

Script generation

To generate the script for the comic book the output of the previous step is given to the large language model Llama 3.1. This model goes through all the data using an adaptive sliding context system. This means that it takes a set number of sentences and then analyses those sentences to see if those can be used to create one panel. The system can respond with 2 different outputs. It has enough data or it needs more data. If the system has enough data it cuts off the data it needs from all the data that it received and uses that to generate a comic book panel script. The data that is not used is feed into the next loop with new data. This means that if the first loop the system receives sentence 0-50 and used 0-38. The next loop uses 38-88. If the system outputs that it needs more data it receives extra sentences, resulting in more context which the model can work with. The usage of this adaptive sliding context system allows for the panel generation to be more coherent and comes with the benefit that there is no lose of data during processing. Besides the sentences the AI can use it also receives a summary of the story so far so that it has more context that it can work with. Context is also why the model did not receive the

entire transcript in one go. As Llama 3.1 can not handle very long text files and will lose understanding of the entire file. Thus this step of giving the model a summary had to be taken so that the model could generate a something that was coherent with the entire process.

One of the limitation of LLama 3.1:8B and many other large language models is the context that it can work with. Llama 3.1 has a context limit of 128k tokens which is about 512000 characters. As we are working with hours of people talking this limit has been exceeded after about 1.5 hours of recording. So to be sure that the system works with any amount of data the input had to be cut up. Furthermore using a lot of context reduces the accuracy and the control over the output. Llama is know to miss a lot of information when working with a lot of context REF HERE. During testing of the Llama for this step it was found that it is rather bad at outputting proper JSON structure and when asked to output very large files of JSON there is a high chance of failure. Because of this the generation of bite sized JSON out put is better as we can now check the output of each step to see if it is proper JSON and if not let the model try again. Because of the nature of LLM's the output is not always correct and thus a test to see if the output is correct had to be done. This testing results in only correct JSON output which can be used in the next step of the pipeline.

The output of this step is a JSON file that you can be seen in listing 4.3.

Listing 4.3: Prompt for Comic Book Script Generation

```
1 You are an AI comic scriptwriter adapting a Dungeons & Dragons session.
2
3 Here is what has happened so far: {story_so_far.strip()}
4
5 Here is the next part of the transcript: {content}
6
7 IMPORTANT FILTERING:
8 - Ignore any rule talk, dice rolls, or out-of-character chatter.
9 - Focus only on in-character dialogue, world-building, and story-driving
   actions.
10 TASK:
11 Generate ONE new panel basepd on a selection of lines from this chunk. Your
   job is to condense meaningful story progression into a single visual
   scene.
12 Output JSON in this format:
13 {{
14 "used_segments": [list of indices],
15 "panel": {{
16 "setting": "...",
17 "characters": [Character names],
18 "objects": [Important objects],
19 "action": "...",
20 "mood": "...",
```

```
21 "camera_view": "...",  
22 "lighting": "...",  
23 "text": "..."  
24 }},  
25 "summary": "One sentence summary of the panel for story memory."
```

Prompt generation

To create an image generation model prompt that is to be used in the next step Llama 3.1 is used again. This time the AI is asked to convert each JSON output of the previous step into a prompt that can be fed into stable diffusion XL. This step is important as most image generative models work well with human language and not a JSON file. Furthermore stable diffusion has one very large limitation as it can only work with 77 tokens. Of which the 2 tokens are reserved as a start and end token. Thus only 75 tokens are available, and that is not a lot. It is unknown how many characters 75 tokens is and how the Stable Diffusion tokenizer work, words can be 1 token, but certain symbols are also 1 token. Because of this Llama 3.1 is asked to create a prompt out of the JSON data which is compatible with the 75 token limit. It is asked to create a visually descriptive SDXL prompt based on the JSON output of the previous step. Furthermore it is given 4 requirements. It should create one natural-language description of the panel image. Use a maximum of 75 tokens. It should not add any other data to the output then the prompt. Lastly it is asked to keep the prompt short and concise so that it sticks within the token limit.

Besides the prompt I have also used the following negative prompt: "deformed, blurry, extra limbs, poorly drawn hands, extra fingers, mutated, out of frame, ugly, tiling, low resolution, bad anatomy, watermark". This negative prompt is used for each image and has it's own token limit. The primary reason for this is to increase the image quality without having to spend tokens in the main prompt, but also to prevent strange generative AI artefacts such as multiple limbs.

Image generation

To generate the images for the comic book the prompt from the previous step is fed into Stable Diffusion XL. Besides this there are only two other things that are given to the model. The negative prompt that was created is given and a LoRA is passed on to create a consistent art style. A LoRA a low-rank adaption of the stable diffusion which modifies the model in a lightweight way so that you do not have to retrain the whole model. It is similar to The LoRA used for this project is the Fantasy Art XL LoRA model from civit.ai. This LoRA is trained to generate comic book from the years 70s and 80s. With this the comic book has a coherent art style without having to spend many prompt tokens on the art style.

Comic book maker

Converting all the images to a comic book structure was done with the pillow python image library. This is a library which can edit images from python code. There are 3 steps to take to create a whole comic book. First the text needs to add the text to the image. This was done by adding a white box at the bottom of the image and placing the text inside the white box. I specifically omitted the usage of speech bubbles here because of two things. It would require an extra AI model which uses computer vision to search for faces so we can get an idee of where the speech bubble needs to point towards, which is extra complexity which is outside of the scope of this thesis. In addition to that I assume that this would create a high chance that the speech bubble would overlap with important parts of the image as it would just be placed at random. Normally speech bubbles are part of the artistic vision of the entire panel. The second step is the combining of the images now with text into one page which was also done with the pillow library. For clarity a black border was placed between the images. Lastly all the images had to be combined into one PDF document which was also done with the pillow library. These 3 steps result in the final resulting comic book.

4.2.2 Whisper

4.2.3 Llama

4.2.4 Image generating models: Stable Diffusion

The chosen image generation model is Stable Diffusion version stabilityai/stable-diffusion-xl-base-1.0 [47]

Chapter 5

Evaluation

5.1 User Case Study

To evaluate the artefact, participants have been asked to fill in a questionnaire. The questionnaire is used to evaluate the artefact on the design requirements and to gather feedback on the artefact. The questionnaire is divided into 6 sections: Background, Representation of the Session, Narrative and Coherence, Visuals and Multimodality, AI Retelling as a Design Challenge, Improvement and Future Use.

Introduction to the Questionnaire

The purpose of this questionnaire is to gather feedback on the comic book generated from your Dungeons and Dragons session. The primary goal of this research is to explore the design requirements for a system that retells tabletop role-playing games, rather than to create a perfect retelling tool. To this end, a prototype comic book generator has been developed using multiple artificial intelligence models.

Because the comic was generated by several AI models, some parts may appear confusing, unexpected, or even inappropriate. Please be aware that your personal character, other characters, dialogue, and artwork may differ from your expectations, and in some cases may include content not suitable for all audiences.

Please read the following questions carefully and provide your responses in the space provided. For multiple-choice questions, select the option that best represents your opinion. For open-ended questions, feel free to elaborate as much as you like. We kindly ask you to answer all questions honestly and to the best of your ability.

All responses will remain confidential, anonymized, and used solely for the research purposes of this master's thesis. If you would like to know more about how the collected data (including questionnaire responses and audio recordings from the sessions) will be used, please feel free to contact me at: rovos@edu.aau.at

Thank you so much for taking the time to participate in my thesis research! Your input means a lot to me, and I truly couldn't complete my studies without your help.

Questionnaire

Introduction

1. Did you participate in one of the Dungeons & Dragons sessions recorded for this project? Yes / No
2. Did you read the comic book before filling out this questionnaire? Yes / No
3. How familiar are you with comic books as a storytelling format? Likert scale (Not at all – Very familiar)

The Three Forms of Play

In tabletop role-playing games, three primary activities typically occur:

- **Exposition:** the Game Master describes the situation and world.
 - **Roleplay:** players act and speak in character, interacting with each other and NPCs.
 - **Rule interaction:** players engage with mechanics such as dice rolls, checks, or adding numbers.
4. To what extent do you feel that the Game Master's exposition is present in the comic book? Likert scale
 5. To what extent do you feel that your in-character roleplay actions are conveyed in the comic book? Likert scale
 6. To what extent do you feel that your actions as a player (choices, decisions, initiatives) are conveyed in the comic book? Likert scale
 7. Did you notice any text or visuals in the comic that explicitly referenced game rules (e.g., dice rolls, checks, modifiers)? Yes / No
 - If yes: About how many times did you notice this? Number box
 8. Would you like rule interactions to be part of the comic book? Yes / No
 9. Of the three activities (exposition, roleplay, rule interaction), which do you feel was represented best in the comic? Why? Open-ended

Ethical Considerations

10. To what extent do you feel that the story told in the comic book still belongs to you and your group, even though it was generated by AI? Likert scale
11. Did the comic include anything you felt was inappropriate, offensive, or misaligned with the spirit of your group's play? Yes / No
 - If yes, please describe what stood out and why. Open-ended
12. Do you believe it is ethically acceptable to use AI systems to retell personal play sessions (such as D&D) in the form of a comic book? Likert scale

Other Reflections

13. Did the comic capture the *feeling* or *atmosphere* of the session for you? Why or why not? Open-ended
14. Was there anything important from the session that you feel was missing or underrepresented in the comic? Please explain. Open-ended
15. How easy was it to follow the story in the comic compared to how you experienced the original session? Likert scale
16. To what extent did the comic capture important aspects of the session (characters, key events, mood)? Likert scale
17. How well did the text and images in the comic work together to retell the session? Likert scale
18. Did the comic feel like a valuable retelling of the session for you and your group? Why or why not? Open-ended
19. If you could improve one aspect of the comic generator, what would it be and why? Open-ended

5.1.1 Results

5.2 Self observation

In addition to participant feedback, I conducted a self-observation of the generated comic book. This allowed me to systematically evaluate the output according to predefined criteria, ensuring consistency and highlighting aspects that participants may not explicitly mention.

5.3 Data analysis

The creation of a comic book from audio with AI involves many moving parts, each of which can be analysed since all intermediate data is saved during processing. We can examine the audio, the transcript, the generated comic book script, the generated prompts, the generated images, the generated text, and finally the comic book as a whole. By analysing each stage, we can identify where the system struggles and where it performs well, which not only informs future improvements but also helps answer the research question. This section begins with an analysis of the transcription stage. The analysis in this chapter is done by looking at the data in a holistic manner: scrolling through everything; reading everything; and making notes on everything, using word searches, counting occurrences of specific words, and looking at distributions of various metrics. Finally ChatGPT-5.0 is consulted to look at the data and give insights that might be valuable, and to make graphs. Everything done with ChatGPT-5.0 is double checked with standard statistic software.

Transcript

The transcription produced by the system went through three processing steps. First, Whisper converted the speech to text. Then, punctuation restoration was performed by FPML, and finally, diarization was carried out by pyannote. Only the final result of this pipeline was saved and analysed in this section. The output of the transcription was a JSON object with four fields: *start*, *end*, *speaker*, and *text*. *Start* marks the time in the audio recording when a speaker begins speaking, and *end* marks when it finishes. *Speaker* identifies the person speaking, and *text* contains the spoken words.

When analysing this data, one issue becomes apparent immediately: the entries are unusually short for full sentences, and the same speaker frequently appears in consecutive entries that should have been merged into a single entry. To illustrate this problem, three plots were generated from the raw dataset. As shown in Figure 5.1, the majority of entries are shorter than 2 seconds. Figure 5.2 shows that most entries contain only one to three words. In addition to short sentences, the text contains artifacts such as excess periods at the end of an entry and stray periods inserted within words. As shown in Figure 5.3, this is quite common, but it also varies substantially between sessions. To strengthen this observation, Figure 5.4 shows no correlation between the number of entries and the number of artifacts. These observations indicate clear diarization errors: the system failed to capture complete sentences as coherent entries. However, since many consecutive entries are assigned to the same speaker, merging them produces a much more accurate representation of the dialogue.

The merged dataset tells a very different story, as shown in Figures 5.5, 5.1, and 5.2. We see a significant reduction in the number of entries, indicating that much of the raw data resulted from over-segmentation. The number of short entries

(1–2 seconds) is noticeably reduced—though they still dominate—while longer entries become more common, and their distribution better reflects the structure of spoken sentences. This highlights how fragmented the raw transcription was and demonstrates the necessity of merging consecutive same-speaker entries to produce usable data.

Considering the possibility that some entries might not be strictly consecutive in the JSON file but still represent a single sentence separated by a short pause or another speaker, an additional test merged entries from the same speaker if they were separated by a small time gap. As shown in Figure 5.6, applying thresholds of 2 seconds and 5 seconds produced substantial changes in the number of entries and brought the session counts much closer together. Furthermore, Figure 5.7 shows that the number of entries per minute becomes very similar between sessions, suggesting a better-structured dataset. As a result, the merged versions are likely to provide higher-quality input for subsequent processing and yield more coherent results when generating the comic book.

Lastly, as shown in Figure 5.8 and Figure 5.9, the distribution of entry durations becomes more spread out after time-based merging. While most entries remain only a few seconds long, the presence of more medium- and longer-length entries suggests the dataset now reflects sentence-level speech more accurately.

There are, of course, other ways to interpret the data (e.g., examining speaker activity, turn-taking, or thematic content). However, these approaches focus more on analysing the session itself than on improving data usability. Because the purpose of this thesis is to create a comic book rather than to analyse play sessions, alternative forms of analysis were deliberately left out in favour of methods that directly enhance comic-book generation.

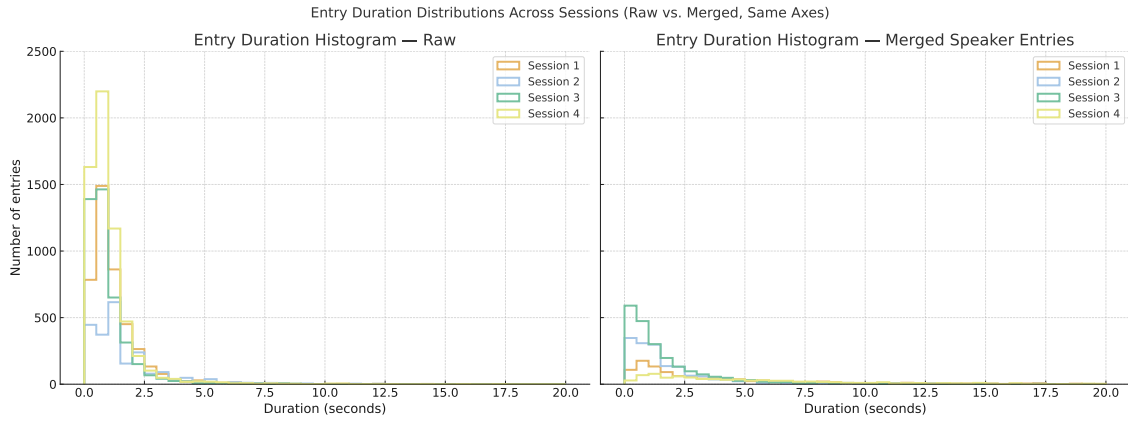


Figure 5.1: Entry duration distributions across all four sessions, comparing raw transcripts (left) with merged speaker entries (right).

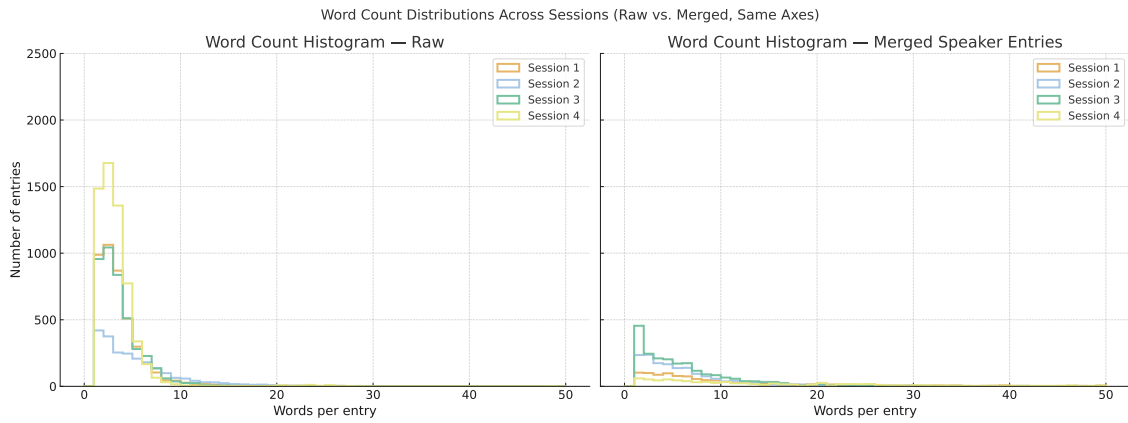


Figure 5.2: Word count distributions across all four sessions, comparing raw transcripts (left) with merged speaker entries (right).

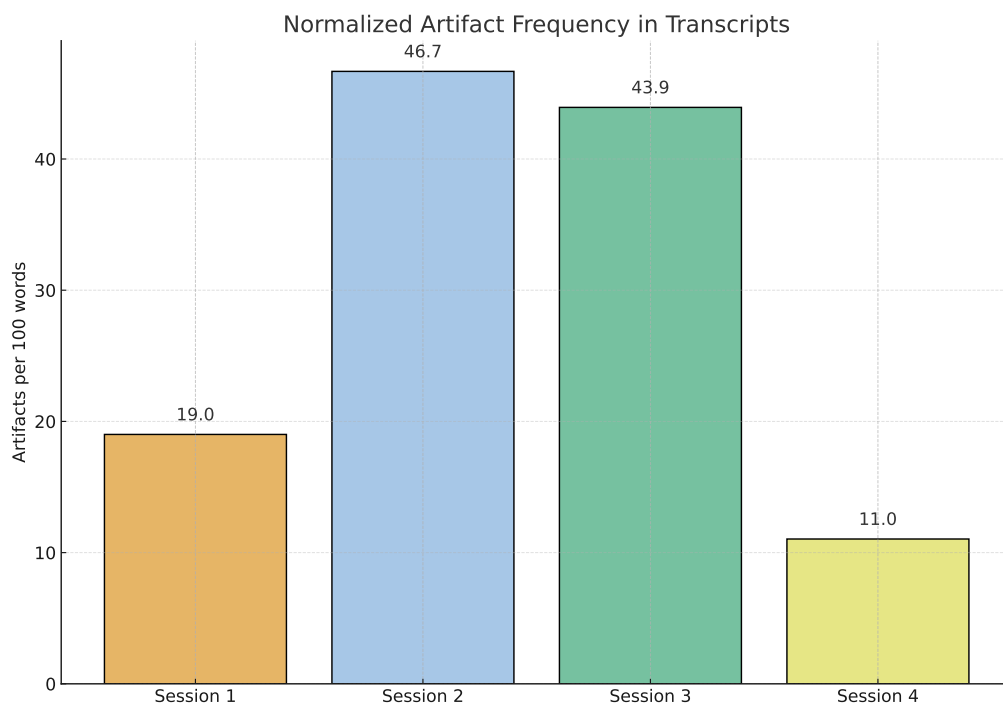


Figure 5.3: Normalized artifact frequency in transcripts, showing artifacts per 100 words for each session.

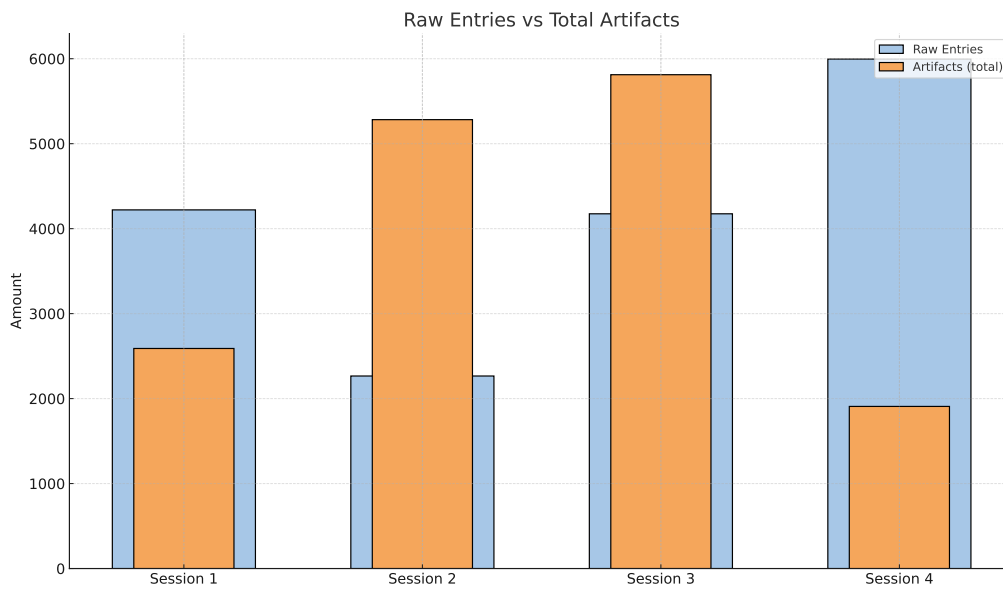


Figure 5.4: Comparison between raw transcript entries and the number of detected artifacts per session.

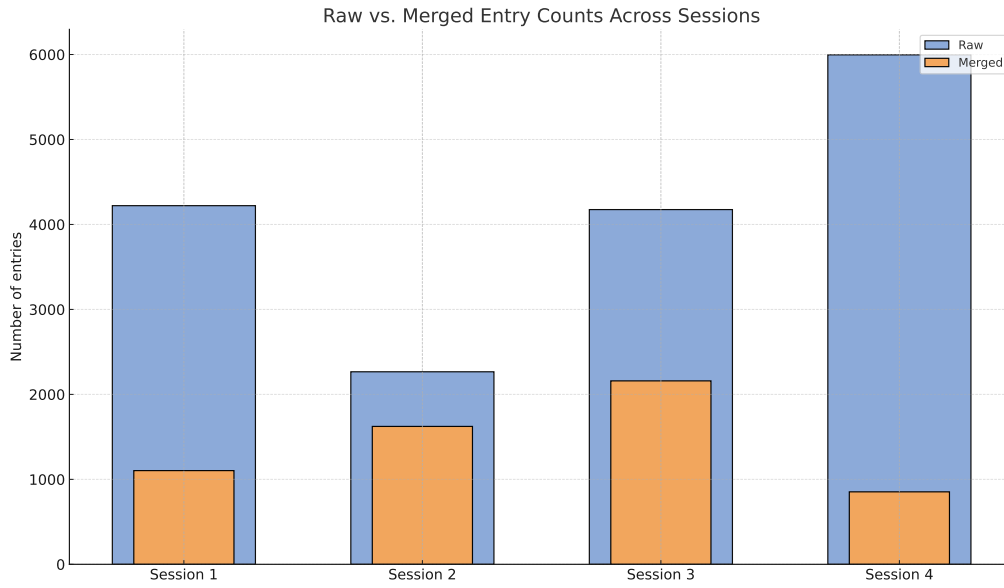


Figure 5.5: Number of transcript entries per session before and after merging consecutive same-speaker segments.

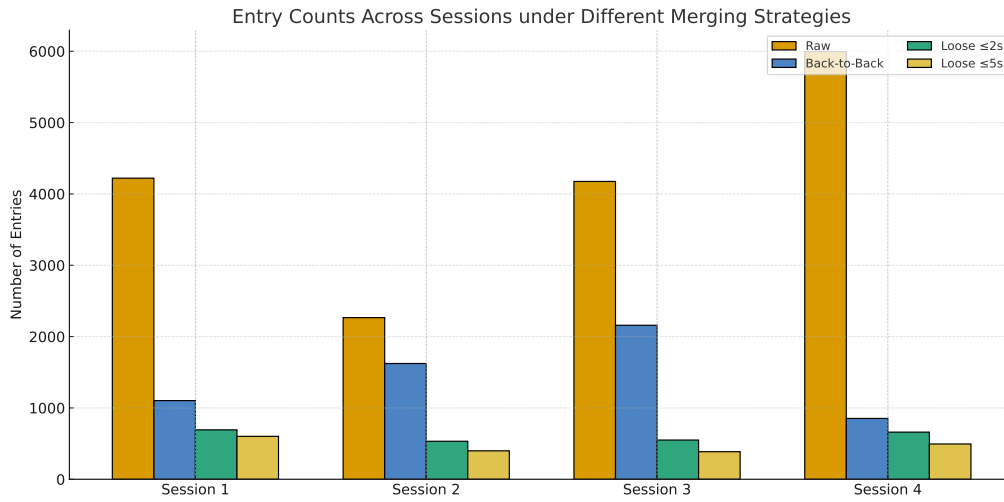


Figure 5.6: Entry counts across sessions under different merging strategies. Bars show the number of transcript entries per session when using raw data, simple back-to-back merging, and looser merges with 2s and 5s thresholds.

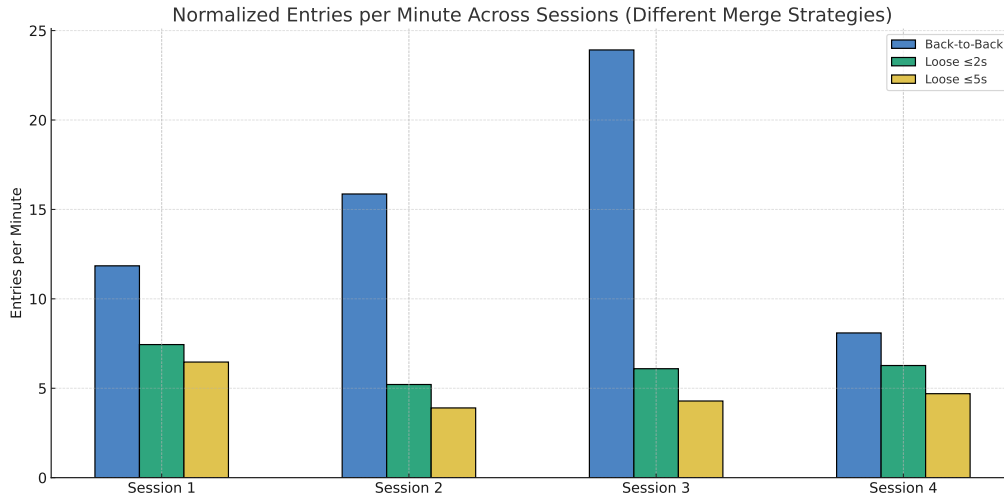


Figure 5.7: Normalized entries per minute across sessions for different merging strategies. Entries are adjusted by total session duration to allow fair comparison.

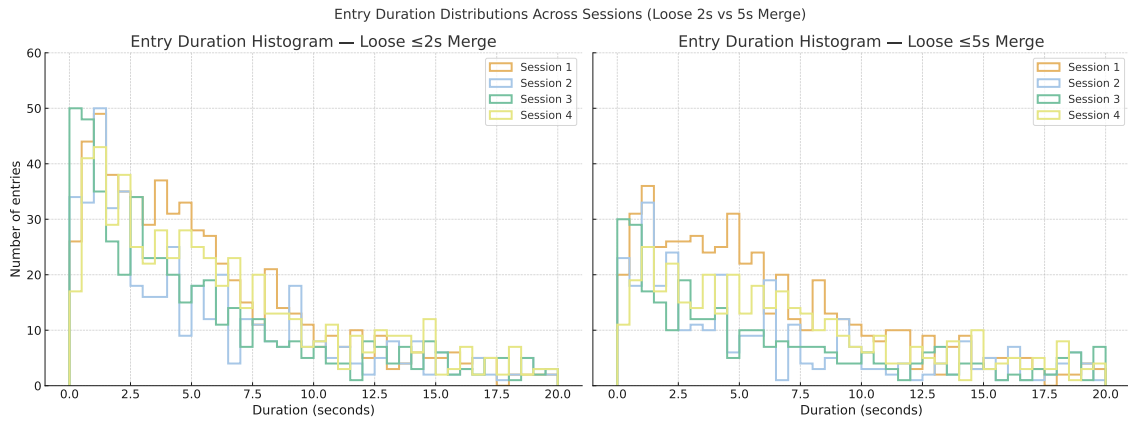


Figure 5.8: Entry duration distributions across all four sessions after time-based merging. Left: loose 2-second merge, Right: loose 5-second merge.

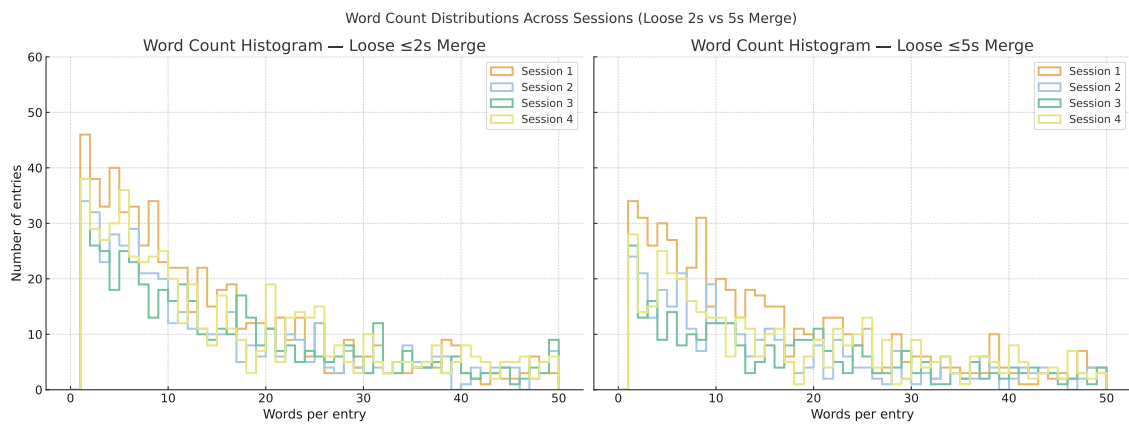


Figure 5.9: Word count distributions across all four sessions after time-based merging. Left: loose 2-second merge, Right: loose 5-second merge.

Script generation

The second step in the pipeline is the creation of the comic book script. This step outputs a JSON file with 8 different keys: **Setting**, **Characters**, **Objects**, **Actions**, **Mood**, **Camera_view**, **Lighting**, and **Text**. These 8 keys are created by Llama 3.1-8B by looking at the output of the transcription in the way described in Chapter ???. The goal is to create enough relevant data which Llama can afterwards use to generate a prompt for the image generation model.

When analysing this data across all four sessions there are a couple of aspects where things are going wrong, but also some things that went well. First, the data structure is always the same, and that is very good as it is the only way for it to be useful in the next step of the pipeline. This consistency is a result of an output check after Llama produces the JSON. However, the **text** field is not always filled, which is a problem for the comic book as it will result in an empty speech bubble. Having a filled **text** key is something that should be added to the check function. Next, the character names are very inconsistent and have strange structures: *Go.nor.r.he* vs. *Gonorrhea* vs. *Gonorrohea* (players tend to give their characters unusual names). But also names like *D.M.*, *Gonorrhea's defeated body*, and *Gonorrhea (in dragon form)* occur occasionally. If you want to create consistency within a product this needs to be cleaned up. It is most likely a result of the transcription process or Llama being confused by strange words that do not link properly to normal language usage.

There is also quite a bit of repetition within the settings, camera, and mood sections. As shown in Figure 5.10 we can see that most camera views generated by the system are *Medium shot*. As shown in Figure 5.11 we can see that most moods are *Tense/Uneasy*, with only a handful of other moods appearing consistently. The “Other” category is actually a diverse collection of moods that did not fit into the chosen bins. However, this aligns with the sessions the players played: they were run in a very tense and fast-paced style where the Dungeon Master pushed the players into quick and dramatic encounters. The same holds true if we look at the grouped settings in Figure 5.12, where we can clearly see that they are in line with what the sessions were about. The “Other” category here primarily consists of mistakes where the system inserted locations that were never part of the adventure, such as *hotel lobby*, *tropical beach*, or *railway corridor*. While not all entries in this category are errors, many are.

The **objects** key is rather inconsistent. Sometimes it gives nice meaningful input like *Chalice of Eternal Life* or *Guard key*. However, it also includes things that are not objects at all: people, descriptions of people, or animals. Even when it does describe the same thing across panels, it does so inconsistently, for example, first it is *cat in the distance*, then *mysterious cat*, and then *a small, unimpressed cat*. There is no clear link between these descriptions other than that they all refer to a cat.

Another aspect that should not be part of the data at this point is gameplay leakage: words like *AC*, *damage*, *roll*, or *check* in the context of rules discussions. A

script was run to count the number of such terms, using the set of keywords ¹. As shown in Table 5.1 and Table 5.2, the percentage of entries with gameplay leakage is small but non-negligible as if any of these appear in the comic book it will break immersion. The system should be improved to avoid this kind of leakage, either by filtering out such entries or by instructing Llama to ignore them.

Dataset	Total Text Entries	With Leakage	% With Leakage
Session 1	292	12	4.1%
Session 2	165	10	6.1%
Session 3	311	11	3.5%
Session 4	456	20	4.4%

Table 5.1: Summary of gameplay mechanics leakage across all four datasets.

Prompt generation

The third step the pipeline is prompt generation. In this step, the structured script data (setting, characters, objects, action, mood, camera view, lighting, and text) is converted into a descriptive natural-language prompt suitable for the image generation model (SDXL 1.0), or at least that is the goal. The purpose of this process is to bridge the gap between structured data and natural language generation. To let the LLM be a creative member of the process and take its place as the creator of the comic book. This step of generating the prompt is both the most creative and also most error-prone step. The JSON keys provide structured data, while the prompt reformulates them into language that SDXL can interpret.

When analysing the generated prompts across all four session they seem quite well. The prompts are very descriptive, they consistently include character names, lighting, shadow, objects, scene setting, location, mood, specific things to showcase, emotions, mood, and more. Each prompt is also very consistent in terms of what they generate and how they generate it the same keys are present each time in the same order. On their own the prompts create a rather vivid description of the situation. However there are also some very obvious artifacts such as the prompt including: "Here is the generated SDXL prompt:" and "/n/n" at the start of the prompt. But also other task descriptive text that should not be part of the prompt. Furthermore many of the prompts are longer then 75 tokens, which is too long for SDXL to work with and thus will cut off part of the end of the prompt. There is no concrete data on this, as there is no tokenizer available for SDXL, but as the images

¹The following mechanics-related keywords were used to detect gameplay leakage: roll, rolled, rolling, dice, die, d20, d12, d10, d8, d6, d4, nat 20, nat 1, natural 20, natural 1, check, skill, saving throw, save, acrobatics, animal handling, arcana, athletics, deception, history, insight, intimidation, investigation, medicine, nature, perception, performance, persuasion, religion, sleight of hand, stealth, survival, initiative, AC (armor class), hit points (HP), damage (dmg), critical (crit), advantage, disadvantage, rule, rules.

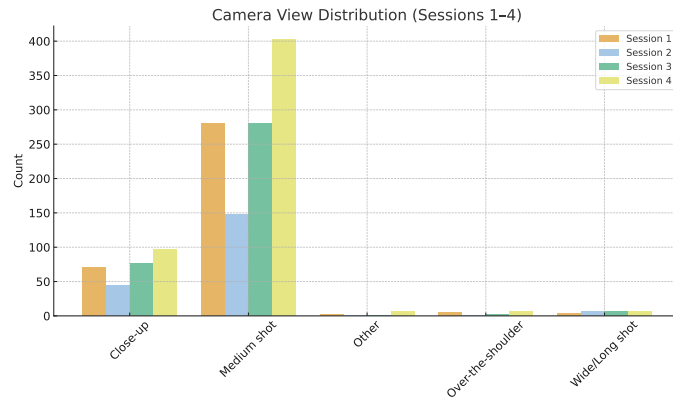


Figure 5.10: Camera view distribution across Sessions 1-4 (grouped into Close-up, Medium shot, Wide/Long shot, Over-the-shoulder, Other).

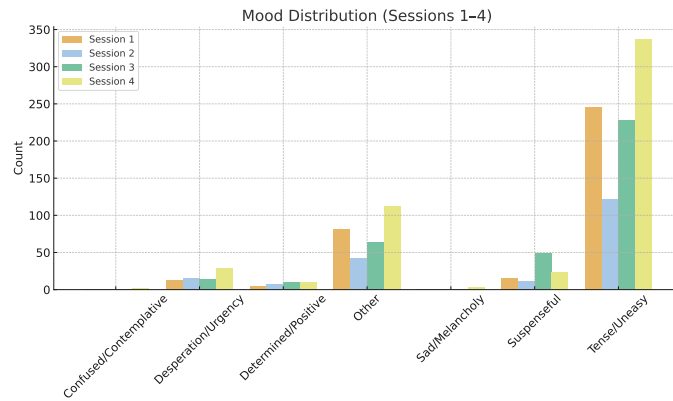


Figure 5.11: Mood distribution across Sessions 1-4 (grouped into Tense/Uneasy, Suspenseful, Desperation/Urgency, Determined/Positive, Sad/Melancholy, Angry/Hostile, Confused/Contemplative, Other).

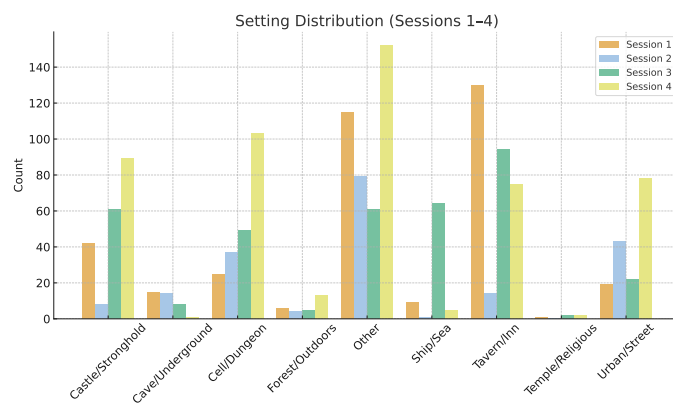


Figure 5.12: Setting distribution across Sessions 1-4 (grouped into Tavern/Inn, Cell/Dungeon, Ship/Sea, Urban/Street, Forest/Outdoors, Temple/Religious, Castle/Stronghold, Cave/Underground, Other).

Term	Session 1	Session 2	Session 3	Session 4	Total
AC	0	0	0	1	1
Advantage	0	2	2	0	4
Animal handling	1	0	0	0	1
Check	3	0	0	4	7
Critical	1	0	1	0	2
Dice mentions	0	0	0	0	0
Damage	0	6	3	5	14
Deception	1	0	0	0	1
Disadvantage	0	1	0	0	1
Hit points	0	0	0	1	1
Initiative	0	0	0	1	1
Investigation	1	0	0	1	2
Medicine	0	0	0	2	2
Nature	0	1	0	0	1
Performance	2	0	0	0	2
Persuasion	2	0	0	0	2
Roll	1	0	2	3	6
Sleight of hand	0	0	1	0	1
Stealth	1	0	1	1	3
Survival	0	0	0	1	1
Total	13	10	10	24	57

Table 5.2: Occurrences of mechanics-related terms in the `text` field across all four datasets, with totals per row and column.

were generated the model said that the prompts were too long and that happened a lot, however no counter was in place to keep track of the data. The errors made during the transcription are also obviously present in the prompts as the same errors reappear in the prompts(Extra dots, weird names, etc.). Besides the quality and errors of each prompt individually there are also issues in-between the prompts and some overarching issues.

When looking at individual prompts they seem relatively usable as there are many consistencies, such as location, characters and objects. However when attempting to understand the story from prompt to prompt it becomes clear that there is no obvious throughline of the story and that almost all prompts have Non-Sequitur transitions[35]. There is no flow, no continuity, no linking of events. Each prompt is a moment in time, without context to the previous or next prompt.

Each prompt makes use of a lot of references to items, objects, location and characters. This is a very good thing, however this data is lost in the image generation as there is no way for SDXL to understand what those things are, there is no database for SDXL to pull from to understand who Arthur is, or what the Chalice of Eternal Life looks like. The model just has to guess based on its training data. This

results in a lot of strange images where characters look nothing like they should, or objects are not what they should be.

5.4 Expert review

5.5 The minimal viable product

The minimal viable product (MVP) is a version of a new product that includes only the essential features necessary to meet the needs for initial testing. The goal of an MVP is to validate the product idea with minimal resources and gather feedback for future development. In figure 5.13 you can see the result of the MVP and in figure 4.1 you can see the pipeline that is used for the MVP.



Figure 5.13: The minimal viable product

5.5.1 The minimal viable product

5.6 Small experiments

5.6.1 The language of TTRPGs

Models like Whisper are trained on a wide variety of audio data, however this is primarily normal language usage. Because of this I anticipate that there will be a large issue with transcribing many fantasy words and terms, thus a small experiment has been done to identify if this is truly a problem. For this experiment 10 fantasy creature names have been recorded and whisper transcribed these 10 names. Only 1 out of the 10 names were transcribed correctly and 2 out of 10 were almost correct 5.3. This inaccuracy is a big issue when transcribing any fantasy game session as it will interfere with many important key moments and key names during a session/campaign. When the group meets a monster called a "Slaad" and Whisper transcribes this as "Nod" it will be quite confusing to understand what is going on in the story.

To generate an accurate transcription of the names a custom model needs to be trained on specific fantasy words and terms, however this is way outside of the scope of this thesis. In addition it is also very difficult as Fantasy names are fantasy names for a reason and people can just come up with new names all the time. Integrating thousands of names however could still be valuable as it will most likely catch most cases.

Correct Name	Transcribed name
Owlbear	Halber
Goblin	Goblin
Rakshasa	Raksasa
Dracolich	Draco Lich
Quasit	Kwasi
Slaad	Nod
Drow	Drought
Illithid	Illidan
Cambion	Kempion
Ankheg	Hank Man

Table 5.3: Comparison of monster names and whisper transcribed versions

5.7 Generating the comic book script

The iterative design process of generating the comic book script after the MVP version was a crucial step to improve the quality of the comic book panels. From

having a one-shot approach where we gave the entire transcript to the LLama 3.1 model, we moved to an adaptive panel generator which has a structured approach to data and keeps track of the context. In this section all the iterative steps are described and what did and did not work in chronological order.

MVP

For the MVP of the artefact the entire transcript of the tabletop role-playing game session was given to Llama 3.1 and see what it would generate. The LLM was asked: "You are given a Dungeons and Dragons session transcript. Convert it into a comic book script." This resulted in two results: The LLM crashed as the token limit was exceeded or it gave things that had nothing to do with the input and hallucinated everything. A quick fix to this was to cut the transcript into smaller chunks of text which resulted in an interesting result as it actually gave a comic book script. It was given 2 scenes of text which were hand cut out of the transcript. However as the LLM was not instructed to give a structured output it was useless as every time the program was run it would give a different output. The last step was to create a structured prompt for the LLM which is the prompt in 4.1. This resulted in a consistent JSON format output which could then be used to generate images.

Cleaning up the transcript

When you feed an audio recording to Whisper it transcribes the audio as one long sentence. There is no punctuation or any idea of sentences structure. This way it is impossible to keep track of who said what and when. To fix this the transcript needs to be cleaned up. The first step is to use a Punctuation and Capitalization Restoration model [25] to add punctuation and capitalization to the text. This will make the text more readable and easier to understand for an LLM. The second step is to use Speaker Diarization to identify the different speakers in the text. The tool used to do this was pyannote 3.1 [7, 42], an open source python library which is super easy to use and broadly used by people (15 million download per month). Lastly combine the cleaned up whisper transcription with the speaker diarization to create a structured JSON file. This new transcript file 5.1 is what created the foundation for the entire comic book generator.

Listing 5.1: Session transcript JSON

```
1 {  
2   "panel_index": 0,  
3   "speaker": "SPEAKER_02",  
4   "text": "Our story actually starts with the three of you...",  
5   "start": 0.0,  
6   "end": 2.96  
7 },  
8 ...
```

Scene detection

Line based chunking

Every line or every X lines

LLM based chunking

Scene cutting

LLM based filtering

Ask the LLM to decide if a line of text is player or character speech. Into giving it a chunk instead.

Context aware LLM based chunking

Adaptive panel generating

5.8 Experimental setup

To test the system on tool functionality, but primarily to test for the design requirements of a retelling system 3 different TTRPG sessions have been recorded and converted to a comic book. These comic books were then handed to the players and feedback was acquired using the INSERTHERE questionnaire. The comic books have also been cross-referenced with each other and with the original recording to see if there are differences in the generated comic book, but which are the same in the sessions.

Experimental sessions

One Dungeons and Dragon 5th edition dungeon master hosted 3 sessions with 2 or 3 players. Each session was the same scenario and lasted 1 to 2 hours. This setup was chosen as this makes the experiment as consistent as possible, but with important possible deviations.

Dungeons and Dragons 5th Edition was chosen as the TTRPG of choice, because there is easy access to many people with a lot of playing experience of this game. It is one of, if not the largest, TTRPG game that is currently being played. It is important to have a game which people know how to play, because I want to minimise the rule based conversations in the recording as much as possible. Rule conversations are an extra dimension of complexity for the system and this should be avoided as much as possible.

Participant feedback

Participant have been asked feedback based on a customized version of the INSERTHERE questionnaire.

5.8.1 Analyzing the generated comic books

Chapter 6

Conclusion

6.1 Summary

6.2 Further Work

6.2.1 Idea 1

The through line of the story is not taken into account.

6.2.2 Idea 2

Multi step image generation + LoRRas - Object permanency

6.2.3 Idea 3

Better models / Custom models VLM

6.2.4 Idea 4

Split audio track

6.2.5 Idea 5

Scene detection

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Appendix A

NetLogo Code

Listing A.1: app.py

```
1 import transcribe
2 import CreateComicPage
3 import TextToScript
4 import scriptToImage
5 import scriptToPrompt
6
7 def main():
8     transcribe.main()
9     TextToScript.main()
10    scriptToPrompt.main()
11    scriptToImage.main()
12    CreateComicPage.main()
13
14 if __name__ == "__main__":
15     main()
```

Listing A.2: transcribe.py

```
1 import whisper
2 import torch
3 from transformers import AutoTokenizer, AutoModelForTokenClassification
4 from pyannote.audio import Pipeline
5 import json
6 import re
7
8 # Load Whisper model
9 whisper_model = whisper.load_model("turbo")
10
11 # Load punctuation restoration model
12 tokenizer = AutoTokenizer.from_pretrained("oliverguhr/fullstop-punctuation-
    multilang-large")
```

```

13 punct_model = AutoModelForTokenClassification.from_pretrained("oliverguhr/
    fullstop-punctuation-multilang-large")
14 punct_model.eval()
15
16 # Load pyannote diarization pipeline (this requires your HuggingFace token)
17 diarization_pipeline = Pipeline.from_pretrained("pyannote/speaker-
    diarization-3.1")
18 diarization_pipeline.to(torch.device("cuda"))
19
20
21 def transcribe_with_segments(audio_path):
22     """Transcribe audio and return timestamped segments."""
23     result = whisper_model.transcribe(audio_path)
24     return result.get("segments", [])
25
26 def restore_punctuation(text):
27     """Restore punctuation and capitalization using the punctuation model."
    ""
28     inputs = tokenizer(text, return_tensors="pt")
29     with torch.no_grad():
30         logits = punct_model(**inputs).logits
31         predictions = torch.argmax(logits, dim=-1)[0].tolist()
32         tokens = tokenizer.convert_ids_to_tokens(inputs['input_ids'][0])
33
34     # Reconstruct text properly handling SentencePiece tokens
35     words = []
36     current_word = ""
37
38     for i, (token, label) in enumerate(zip(tokens, predictions)):
39         # Skip special tokens
40         if token in ['<s>', '</s>', '<pad>', '<unk>']:
41             continue
42
43         # Handle SentencePiece tokens (they start with _which is \u2581)
44         if token.startswith('_') or token.startswith('\u2581'):
45             # Start of new word
46             if current_word:
47                 words.append(current_word)
48             current_word = token.replace('_', '').replace('\u2581', '')
49         elif token.startswith('##'):
50             # BERT-style subword continuation
51             current_word += token[2:]
52         else:
53             # Regular token or continuation
54             if current_word and not token.startswith('_') and not token.
                startswith('\u2581'):

```

```

55         current_word += token
56     else:
57         if current_word:
58             words.append(current_word)
59             current_word = token.replace('_', '').replace('\u2581', '')
60
61     # Add punctuation based on label
62     if label == 1: # PERIOD
63         current_word += "."
64     elif label == 2: # COMMA
65         current_word += ","
66     elif label == 3: # QUESTION_MARK
67         current_word += "?"
68
69     # Don't forget the last word
70     if current_word:
71         words.append(current_word)
72
73     # Join words and clean up
74     punctuated_text = " ".join(words)
75
76     # Clean up any remaining special characters and normalize spaces
77     punctuated_text = re.sub(r'[_\u2581]', '', punctuated_text)
78     punctuated_text = re.sub(r'\s+', ' ', punctuated_text)
79     punctuated_text = punctuated_text.strip()
80
81     # Capitalize first letter
82     if punctuated_text:
83         punctuated_text = punctuated_text[0].upper() + punctuated_text[1:]
84
85     return punctuated_text
86
87 def diarize_audio(audio_path):
88     """Run diarization and return speaker segments with timestamps."""
89     diarization = diarization_pipeline(audio_path)
90     diarized_segments = []
91     for turn, _, speaker in diarization.itertracks(yield_label=True):
92         diarized_segments.append({
93             "start": turn.start,
94             "end": turn.end,
95             "speaker": speaker
96         })
97     return diarized_segments
98
99 def merge_transcript_and_diarization(transcript_segments, diarized_segments
):

```

```

100     """Match each transcript segment to a speaker based on overlap."""
101     merged = []
102
103     for t_seg in transcript_segments:
104         # Find diarization segments overlapping this transcript segment
105         candidates = [d for d in diarized_segments if not (d["end"] < t_seg[
106             "start"] or d["start"] > t_seg["end"])]
107
108         if candidates:
109             # Pick the speaker with the longest overlap
110             best_speaker = None
111             max_overlap = 0
112             for c in candidates:
113                 overlap_start = max(t_seg["start"], c["start"])
114                 overlap_end = min(t_seg["end"], c["end"])
115                 overlap = overlap_end - overlap_start
116                 if overlap > max_overlap:
117                     max_overlap = overlap
118                     best_speaker = c["speaker"]
119             else:
120                 best_speaker = "Unknown"
121
122         merged.append({
123             "start": t_seg["start"],
124             "end": t_seg["end"],
125             "speaker": best_speaker,
126             "text": t_seg["text"]
127         })
128
129     return merged
130
131 def create_clean_transcript_with_speakers(audio_path, output_json):
132     print("Transcribing audio with Whisper...")
133     transcript_segments = transcribe_with_segments(audio_path)
134
135     print("Restoring punctuation...")
136     for seg in transcript_segments:
137         seg["text"] = restore_punctuation(seg["text"])
138
139     print("Running speaker diarization...")
140     diarized_segments = diarize_audio(audio_path)
141
142     print("Merging transcript and diarization...")
143     merged = merge_transcript_and_diarization(transcript_segments,
144         diarized_segments)

```

```

144     with open(output_json, "w", encoding="utf-8") as f:
145         json.dump(merged, f, indent=2, ensure_ascii=False)
146
147     print(f"Transcript with speakers saved to {output_json}")
148
149 def main():
150     audio_file = "../audio/fullSession2.wav"
151     output_file = "output/clean_transcript_with_speakers.json"
152     create_clean_transcript_with_speakers(audio_file, output_file)
153
154 if __name__ == "__main__":
155     main()

```

Listing A.3: TextToScript.py

```

1 import ollama
2 import json
3 import re
4 import time
5
6 def load_transcript(path):
7     with open(path, "r", encoding="utf-8") as f:
8         return json.load(f)
9
10 def adaptive_panel_generator(transcript, chunk_size=15, max_retries=3):
11     panels = []
12     position = 0
13     memory_buffer = []
14     story_so_far = ""
15
16     while position < len(transcript):
17         chunk = transcript[position : position + chunk_size]
18         print(f"Processing chunk at line {position}...")
19
20         retries = 0
21         while retries < max_retries:
22             try:
23                 used_indices, panel, summary = generate_panel_from_chunk(
24                     chunk, story_so_far)
25                 if panel:
26                     panels.append(panel)
27
28                 memory_buffer.append(summary)
29                 if len(memory_buffer) > 10:
30                     memory_buffer.pop(0)
31                 story_so_far = "\n".join(memory_buffer)

```

```

31
32         num_used = max(used_indices) + 1 if used_indices else 1
33         position += num_used
34         break # success, break out of retry loop
35     else:
36         retries += 1
37         print(f"No panel returned, retrying... ({retries}/{
38             max_retries})")
39         time.sleep(0.3)
40     except Exception as e:
41         retries += 1
42         print(f"Panel generation error: {e} (retry {retries}/{
43             max_retries})")
44         time.sleep(0.3)
45
46     else:
47         print("[x]Failed after maximum retries, skipping chunk.")
48         position += 1 # give up and move on
49
50     return panels
51
52 def generate_panel_from_chunk(chunk, story_so_far):
53     content = "\n".join(f"{i}: {seg['speaker']}: {seg['text']}" for i, seg
54         in enumerate(chunk))
55
56     prompt = f"""
57     You are an AI comic scriptwriter adapting a Dungeons & Dragons session.
58
59     Here is what has happened so far:
60     {story_so_far.strip()}
61
62     Here is the next part of the transcript:
63     {content}
64
65     IMPORTANT FILTERING:
66
67     - Ignore any rule talk, dice rolls, or out-of-character chatter.
68     - Focus only on in-character dialogue, world-building, and story-driving
69       actions.
70
71     TASK:
72
73     Generate ONE new panel based on a selection of lines from this chunk. Your
74     job is to condense meaningful story progression into a single visual
75     scene.

```

```

71 Output JSON in this format:
72 {{
73     "used_segments": [list of indices],
74     "panel": {{
75         "setting": "...",
76         "characters": [Character names],
77         "objects": [Important objects],
78         "action": "...",
79         "mood": "...",
80         "camera_view": "...",
81         "lighting": "...",
82         "text": "..."
83     }},
84     "summary": "One sentence summary of the panel for story memory."
85 }}
86
87 - Every panel must include all 9 fields and cannot be empty.
88 - Output only valid JSON -no extra text, no commentary.
89 """
90
91 response = ollama.chat(
92     model="llama3.1",
93     messages=[{"role": "user", "content": prompt}],
94     stream=False,
95 )
96
97 output = response["message"]["content"]
98 return parse_adaptive_json(output)
99
100 def parse_adaptive_json(text):
101     try:
102         match = re.search(r'{{.*}}', text, re.DOTALL)
103         if not match:
104             raise ValueError("No JSON found")
105         json_text = match.group(0)
106         json_text = re.sub(r",\s*}", "}", json_text)
107         json_text = re.sub(r",\s*\]", "]", json_text)
108         parsed = json.loads(json_text)
109
110         used = parsed.get("used_segments", [])
111         panel = parsed.get("panel", {})
112         summary = parsed.get("summary", "")
113
114         return used, panel, summary
115     except Exception as e:
116         print("JSON parsing failed:", e)

```

```

117         return [], None, ""
118
119 def save_panels(panels, output_path):
120     with open(output_path, "w", encoding="utf-8") as f:
121         json.dump(panels, f, indent=2, ensure_ascii=False)
122
123 def main():
124     transcript_path = "output/clean_transcript_with_speakers.json"
125     output_path = "output/adaptive_comic_script.json"
126
127     transcript = load_transcript(transcript_path)
128     panels = adaptive_panel_generator(transcript)
129     save_panels(panels, output_path)
130     print(f"[check] Script saved to {output_path}")
131
132 if __name__ == "__main__":
133     main()

```

Listing A.4: scriptToPrompt.py

```

1 import json
2 import ollama
3 import time
4
5 def normalize_panel(panel):
6     def normalize_entry(entry):
7         if isinstance(entry, str):
8             return entry
9         elif isinstance(entry, dict):
10            name = entry.get("name", "Unknown")
11            extras = [v for k, v in entry.items() if k != "name" and v]
12            return f"{name} ({', '.join(extras)})" if extras else name
13        return str(entry)
14
15    panel["characters"] = [normalize_entry(c) for c in panel.get("
        characters", [])]
16    panel["objects"] = [normalize_entry(o) for o in panel.get("objects",
        [])]
17
18    return panel
19
20 def generate_prompt_with_llama(panel):
21     system_message = "You are a prompt engineer for Stable Diffusion XL (
        SDXL)."
22
23     user_message = f"""

```

```

24 Create a visually descriptive SDXL prompt based on the following panel
    description:
25
26 Setting: {panel.get('setting', '')}
27 Characters: {"", ".join(panel.get("characters", []))}
28 Objects: {"", ".join(panel.get("objects", []))}
29 Action: {panel.get('action', '')}
30 Mood: {panel.get('mood', '')}
31 Camera View: {panel.get('camera_view', '')}
32 Lighting: {panel.get('lighting', '')}
33
34 Requirements:
35 - Combine all of the above into a single natural-language description.
36 - Maximum length: 75 tokens.
37 - Do NOT add quotes or commentary, just return the prompt text only.
38 - The prompts need to be short, concise, and suitable for generating a
    single comic book panel.
39 """
40
41 response = ollama.chat(
42     model="llama3.1",
43     messages=[
44         {"role": "system", "content": system_message},
45         {"role": "user", "content": user_message}
46     ],
47     stream=False
48 )
49
50 prompt = response["message"]["content"].strip()
51 return prompt
52
53 def process_script(input_path, output_path):
54     with open(input_path, "r", encoding="utf-8") as f:
55         script = json.load(f)
56
57     for i, panel in enumerate(script):
58         print(f"Generating SDXL prompt for panel {i}...")
59         try:
60             panel = normalize_panel(panel)
61             prompt = generate_prompt_with_llama(panel)
62             panel["sdxl_prompt"] = prompt
63             panel["negative_prompt"] = (
64                 "deformed, blurry, extra limbs, poorly drawn hands, extra
                    fingers, "
65                 "mutated, out of frame, ugly, tiling, low resolution, bad
                    anatomy, watermark"

```

```

66         )
67     except Exception as e:
68         print(f"Failed for panel {i}: {e}")
69         panel["sdxl_prompt"] = ""
70         panel["negative_prompt"] = ""
71         time.sleep(0.3)
72
73     with open(output_path, "w", encoding="utf-8") as f:
74         json.dump(script, f, indent=2, ensure_ascii=False)
75
76     print(f"SDXL prompts saved to {output_path}")
77
78 def main():
79     process_script(
80         input_path="output/adaptive_comic_script.json",
81         output_path="output/adaptive_comic_script_with_prompts.json"
82     )
83
84 if __name__ == "__main__":
85     main()

```

Listing A.5: scriptToImage.py

```

1 from diffusers import DiffusionPipeline
2 import torch
3 import AddTextToImage
4 import json
5
6 testing = False
7
8 def generate_images_from_script(script_path):
9     # Load comic panel script with LLM-generated prompts
10    with open(script_path, "r", encoding="utf-8") as f:
11        comic_script = json.load(f)
12
13    # Load SDXL pipeline
14    pipe = DiffusionPipeline.from_pretrained(
15        "stabilityai/stable-diffusion-xl-base-1.0",
16        torch_dtype=torch.float16,
17        use_safetensors=True,
18        variant="fp16"
19    )
20    pipe.load_lora_weights("../lora/Fantasy_art_XL_V1.safetensors")
21    pipe.to("cuda")
22
23    for i, panel in enumerate(comic_script):

```

```

24     prompt = "Comic book illustration. Vivid fantasy." + panel.get("
        sdxl_prompt", "")
25     negative_prompt = panel.get("negative_prompt", "")
26
27     if not prompt:
28         print(f"Skipping panel {i}: no prompt found.")
29         continue
30
31     image = pipe(
32         prompt=prompt,
33         negative_prompt=negative_prompt,
34     ).images[0]
35
36     image.save(f"output/images/imagesWithoutText/panel_{i}.png")
37     image_with_text = AddTextToImage.add_text_to_panel(panel["text"],
        image)
38     image_with_text.save(f"output/images/imagesWithText/panel_{i}.png")
39
40     if i == 5 and testing:
41         return
42
43 def main():
44     generate_images_from_script("output/adaptive_comic_script_with_prompts.
        json")
45
46 if __name__ == "__main__":
47     main()

```

Listing A.6: CreateComicPage.py

```

1 from PIL import Image
2 import os
3
4 def resize_and_add_border(image, target_size, border_size):
5     resized_image = Image.new("RGB", target_size, "black")
6     resized_image.paste(image, ((target_size[0] - image.width) // 2, (
        target_size[1] - image.height) // 2))
7     return resized_image
8
9 def load_images():
10     number_of_images = 6
11     images_loaded = []
12     image_files = os.listdir("output/images/imagesWithText")
13
14     # Sort filenames by panel index (assumes filenames like panel_0.png)
15     sorted_files = sorted(

```

```

16     image_files,
17     key=lambda name: int("".join(filter(str.isdigit, name)))
18 )
19
20 for i in range(len(sorted_files)):
21     image_path = os.path.join("output/images/imagesWithText",
22                               sorted_files[i])
23     image = Image.open(image_path)
24     images_loaded.append(image)
25
26     return images_loaded
27
28 def main():
29     images = load_images()
30     columns, rows = 2, 3
31     images_per_page = columns * rows
32
33     for i in range(0, len(images), images_per_page):
34         page_images = images[i:i + images_per_page]
35
36         # Calculate the size of the output image
37         output_width = columns * page_images[0].width + (columns - 1) * 10
38         output_height = rows * page_images[0].height + (rows - 1) * 10
39
40         result_image = Image.new("RGB", (output_width, output_height), "
41             black")
42
43         # Combine images into a grid with black borders
44         for j, img in enumerate(page_images):
45             x = (j % columns) * (img.width + 10)
46             y = (j // columns) * (img.height + 10)
47
48             resized_img = resize_and_add_border(img, (page_images[0].width,
49                                                         page_images[0].height), 10)
50             result_image.paste(resized_img, (x, y))
51
52         result_image = result_image.resize((1024, 1536))
53         result_image.save(f"comicPages/comicPage_{i // images_per_page}.jpg"
54 )
55         print(f"Saved comic page: comicPage_{i // images_per_page}.jpg")
56
57 if __name__ == "__main__":
58     main()

```

Listing A.7: pdfMerger.py

```
1 from PIL import Image
2 import os
3
4 image_dir = "comicPages"
5 image_files = [f for f in os.listdir(image_dir) if f.lower().endswith((".
    jpg", ".jpeg", ".png"))]
6 image_files.sort()
7
8 images = []
9 for file in image_files:
10     img_path = os.path.join(image_dir, file)
11     img = Image.open(img_path).convert("RGB")
12     images.append(img)
13
14 if images:
15     images[0].save("comic.pdf", save_all=True, append_images=images[1:])
```